

MACE: Masked-Autoencoder Conditional Excess for scalable, self-validating detection of driven variables in large dynamical systems

Akanda Wahid-Ul-Ashraf

August 15, 2026 — preprint draft

Licensed CC BY 4.0. Code, experiment ledger, pre-registration documents and figure sources:

<https://github.com/AkandaAshraf/DeepFeatSelection>

Abstract

We address the problem of determining which variables of a large dynamical system are driven by the system and which evolve autonomously. Existing methods cannot answer this at scale: we measured full pairwise convergent cross mapping at $\approx 15,800$ compute-hours for one scan of 10^4 variables, k -nearest-neighbour conditioning became sample-starved by 18 conditioning dimensions, and PCMRI with partial correlation produced mediated false positives near synchrony. We first isolate three mechanisms by which *learned* readouts fail on such systems, each with a separating experimental signature: maturity collapse of difference-based importance under Takens redundancy, typicality substitution in masked-reconstruction scores, and the inability of ensembling to remove shared bias. Together they imply a corrected statistic, an amortised Granger/transfer-entropy-type quantity we call *Masked-Autoencoder Conditional Excess* (MACE): the gain in one-step predictability of a variable when a learned low-dimensional code of the entire remaining system is added to a flexible model of the variable’s own history. The code is shared across variables and each readout is a ridge regression, so a complete scan of 71,721 variables runs in minutes on a consumer laptop, four orders of magnitude below the pairwise equivalent. We give the statistic a theoretical account, not an empirical one alone: its population value is exactly the fraction of a variable’s next-step variance that its own history cannot explain, which is zero precisely for autonomous variables, and both the linear cost and the null of the embedded *ghost channel* (a circularly shifted copy whose score must be zero, giving every dataset a built-in falsification test) follow as corollaries. The argument also corrects our own procedure, since a ghost’s guarantee holds only when its donor’s self-baseline saturates. MACE attains top-10 precision 30/30 on synthetic systems with known membership. Across four open real-world domains it behaves as theory predicts and fails informatively where it does not: it recovers the *C. elegans* motor-command circuit without anatomical labels and tracks a genetic silencing of the command hub AVA at the level of the affected module; a 71,721-neuron zebrafish driven core is spatially coherent; seizures concentrate drive onto fewer channels without increasing it, and per-event maps do not replicate; and in 77 years of sea-level pressure the driven core is the deep tropics while an embedded solar-forcing channel scores zero. We state the limits explicitly: drivenness only, so sources are invisible by design; precision 0.95 at 28% recall, so absence from the driven core is not evidence of autonomy; and validity established down to $n \approx 2,000$ samples. We release code, pre-registrations, and a ledger including all negative results.

Highlights

- A scalable statistic, excess-over-self predictability, that identifies which variables of a large dynamical system are driven by the system.
- Three mechanisms by which learned causal readouts fail on redundant systems, each isolated with a separating experimental signature.
- An embedded negative control, the ghost channel, giving every scan a falsification test that requires no ground truth.
- A theoretical account of why the statistic works: its population value is the share of a variable’s next-step variance its own history cannot explain, the ghost’s null is a corollary rather than a calibration, and the linear cost in V follows from the same embedding argument.
- Whole-brain scans of 71,721 neurons in minutes on a single consumer laptop, where a pairwise equivalent would need $\sim 800,000$ GPU-hours.
- Validation across four open domains, including an interventional test in which the statistic registers a targeted genetic silencing in the pre-registered direction, with per-cell nulls reported.

Keywords: causal discovery; time series; dynamical systems; drivenness; masked autoencoders; transfer entropy; surrogate data; whole-brain imaging; climate teleconnections

1 Introduction

Causal structure discovery from time series is well served by established tools at small scale. Convergent cross mapping exploits Takens’ embedding theorem to detect coupling between pairs of variables on shared attractors [1, 2, 44]; constraint-based frameworks such as PCMCI (the Peter–Clark, or PC, algorithm combined with a Momentary Conditional Independence test) condition away spurious associations across moderate variable counts [4, 26]; and Granger causality and transfer entropy define the underlying conditional-predictability quantity [5, 6]. The question increasingly posed in practice, however, is which of ten thousand observed variables the system drives. That question lies outside the demonstrated operating range of every existing tool. Neural implementations of Granger causality report tens to ~ 100 variables [7], and amortised causal discovery more broadly remains demonstrated at comparable scale [46]; our own wall-clock measurement of pairwise CCM under the full protocol (~ 1.1 s per unordered pair at $n=2,000$, both directions returned by one call) projects a single 10^4 -variable scan at roughly 15,800 compute-hours (≈ 1.8 years); and the natural accelerations fail in our tests (cross-mapping against principal-component summaries scores its own control at the maximum; k NN conditioning [30] collapses at 18 conditioning dimensions in a two-seed synthetic test).

No existing tool closes this. Our own measurements above leave pairwise CCM wall-clock infeasible at this width, and its natural acceleration via k NN conditioning [30] sample-starved by 18 dimensions; PCMCI [4] is untested at this width and, on the synthetic chains examined later in this paper (Section 9), acquires mediated false positives near synchrony rather than degrading gracefully; and the closest learned relatives, neural Granger causality [7] and amortised causal discovery more broadly [46], stop at tens to ~ 100 variables and ship no embedded, ground-truth-free validity check of their own. **No existing method estimates variable-level drivenness**

at $V \gtrsim 10^3$ with a built-in, ground-truth-free validity check; this is the gap this paper closes.

This paper closes that gap. What is new is not the causal quantity: Granger causality and transfer entropy [5, 6, 29] already define what excess-over-self predictability measures, masked autoencoders [9, 47] are an existing representation-learning tool rather than one introduced here, and the ghost channel is a direct descendant of surrogate-data testing [22]. The novelty is in what makes that existing quantity usable at $V \gtrsim 10^3$, in three parts. **(i)** We isolate, with separating experimental signatures rather than an aggregate accuracy drop, three distinct mechanisms by which learned causal readouts fail on redundant dynamical systems: maturity collapse of difference-based importance, typicality substitution in masked reconstruction, and ensembling’s inability to remove shared bias (Section 3). **(ii)** These mechanisms jointly dictate, rather than merely motivate, a single corrected statistic: an amortised Granger/transfer-entropy-type quantity whose self-baseline cancels typicality by construction and whose embedded ghost channel supplies the built-in, ground-truth-free validity check that closes the gap above. The statistic is amortised, via unsupervised compression, to the tens of thousands of variables demonstrated later in this paper, where the classical and neural incumbents above are wall-clock infeasible, sample-starved, mediation-prone, or undemonstrated (Section 4); we validate it where membership is known by construction (Section 7). **(iii)** Beyond synthetic benchmarks, we deploy the statistic across four open real-world domains at a cost small enough to place the method within reach of any laboratory with a workstation, and subject it to a stronger evidence class than correlational validation alone: an interventional test in which the statistic, computed purely observationally, registers a targeted genetic silencing of the *C. elegans* command hub AVA at the level of the affected module, in directions stated before the data was opened and with per-cell nulls reported, alongside a measured method-selection map (Sections 8–9). The rest of this paper is structured as follows. Section 3 isolates the three failure mechanisms. Section 4 derives the proposed method from them and states its cost. Section 7 validates it on systems whose driven set is known by construction, and Section 8 deploys it across four open real-world domains, including an interventional test. Section 9 distils an empirical method-selection map, Section 11 states the limitations, and Section 12 concludes.

Nomenclature

This work spans dynamical systems, machine learning, neuroscience, clinical electrophysiology and climate science; we define the recurring terms once, here. *Driven / drivenness*: a variable is driven if the state of the rest of the system carries information about its next value beyond the variable’s own history; its degree is the excess statistic of Eq. 1. *Autonomous*: evolving under its own dynamics only; the opposite of driven. *Source (exogenous root)*: a variable that influences others but is driven by nothing observed; invisible to the statistic by design. *Delay embedding*: the vector of a variable’s recent lagged values, which for deterministic systems reconstructs the underlying state [2]. *Ghost channel*: a circularly time-shifted copy of a real variable, embedded in the analysis as a negative control. *Masked autoencoder*: a neural network trained to restore randomly hidden inputs from the remainder, yielding a compressed *code* of the whole system. *Ensemble / consensus*: several identically trained networks differing only in random initialisation, and the average of their readouts. *Self-baseline*: the predictability of a variable from its own history alone, subtracted in Eq. 1. *Top- k precision*: the fraction of the k highest-ranked variables that are true positives. *Average precision (AP)*: the area under the precision-recall curve; its chance level equals the positive-class fraction (the *baseline*). *AUROC*: area under the

receiver-operating-characteristic curve; 0.5 is chance. *Ictal / interictal*: during / between epileptic seizures. *Connectome*: the anatomical wiring diagram of a nervous system. *Reanalysis*: a historical gridded reconstruction of the atmosphere from observations. *Teleconnection*: a statistical linkage between climate variability in distant regions. *Pre-registration*: fixing predictions and analysis choices in a written artifact before inspecting the data.

Registration discipline varied by study and we state it exactly: the EEG and climate deployments were registered in a protocol document before their data was accessed; the wild-type worm predictions were fixed in the analysis script’s header before execution, with two animals reserved untouched for held-out confirmation; the intervention directions were stated in the working ledger before the silenced-worm data was opened; the zebrafish analysis was exploratory apart from its in-code ghost gate. Negatives are reported at the same volume as positives, and every number below is reproducible from the released code and public data.

2 Related work

This work draws on five literatures and is validated against a sixth. We survey each in the order in which it enters the paper, and state at the end exactly which of them MACE belongs to.

State-space causal discovery. Convergent cross mapping (CCM) asks whether one variable’s delay-embedded attractor can predict another’s: if X drives Y , then Y ’s reconstructed state retains X ’s imprint and can cross-map it, a consequence of Takens’ embedding theorem [1, 2]. Later work in the same framework separates time-delayed from contemporaneous interactions by scanning the cross-map over lags [44]. The theorem that licenses cross mapping is also the source of the redundancy problem analysed in Section 3: a driver’s imprint is recoverable from any one of its effects, so no single effect is necessary to recover it. These methods are unconditional and pairwise. A scan of V variables requires a cross-map fit per pair, so cost grows with V^2 , and the constant is not small: we measured ~ 1.1 s per unordered pair at $n=2,000$ under the full protocol (one call returns both directions), which projects one scan of 10^4 variables at roughly 15,800 compute-hours. This is the cost consequence this paper measures rather than assumes, and it is the reason the method-selection map of Section 9 recommends CCM at small width (where it detects coupling 0.01 in seconds) and not at large. The natural accelerations failed in our tests: cross-mapping against principal-component summaries scores its own control at the maximum. One feature of CCM is directly ancestral to our statistic, and we note it here rather than claim it: its convergence-with-library-size criterion is an implicit control against a variable’s own predictability, which Eq. 1 makes explicit and cheap.

Constraint-based and information-theoretic approaches. PCMCI pairs a Peter–Clark condition-selection step with a momentary-conditional-independence test, and is built to suppress the false positives that autocorrelation and common drivers produce across moderate variable counts [4, 26]. It estimates, by conditional independence testing, the quantity that Granger causality defines by prediction: X is causal for Y if X ’s past improves prediction of Y beyond Y ’s own past [5]. Transfer entropy is the model-free information-theoretic form of the same quantity [6, 41], and the two coincide exactly for Gaussian variables [29]. In practice the conditional dependence is estimated either parametrically (partial correlation, as in the PCMCI variant we ran) or with nearest-neighbour estimators [30], and the family as a whole admits a compressibility reading, in which the causal factorisation is the one that describes the joint process most concisely [45]. *MACE belongs to this family rather than competing with it.* Eq. 1 is a conditional predictability gain over the target’s own history, which is Granger’s definition, with

two substitutions: the conditioning set is a learned low-dimensional code of the whole remaining system rather than an explicitly selected set of parent candidates, and the estimator is a ridge readout rather than an independence test. Both substitutions trade resolution for width, and the trade is what the measured limits of the incumbents make necessary here: k NN conditioning became sample-starved at 18 conditioning dimensions in our two-seed synthetic tests, and PCMCI is untested at $V \gtrsim 10^3$ and, on the synthetic chains of Section 9, acquires mediated false positives near synchrony rather than degrading gracefully. On the real datasets where both PCMCI and CCM were affordable we measured no separation between them (Section 9), and the map recommends them side by side accordingly.

Learned and amortised causal discovery. Neural Granger causality replaces the linear vector autoregression of the classical test with neural predictors and structured sparsity on the input weights, recovering Granger structure under nonlinear dynamics; the demonstrations report tens to ~ 100 variables [7]. Amortised causal discovery trains a shared inference model so that a graph is produced by a forward pass instead of a per-dataset search, and is likewise demonstrated at comparable scale [46]. MACE takes the amortisation idea and applies it one level down: the encoder is trained once per system, unsupervised, and the per-variable query is a ridge fit, so the cost of asking about one more variable is milliseconds rather than a refit. Two differences from that literature should be stated plainly. First, the target is weaker, and deliberately so: those methods estimate edges, whereas MACE estimates a single scalar per variable. That scalar is the object that remains estimable at the widths of Section 8, and edge-level maps on dense real circuits were recovered by no method we tested, ours included (Section 9). Second, we are aware of no embedded, ground-truth-free validity check in either line of work: a practitioner running such a method on a novel recording obtains a graph, but no quantity within that output whose value certifies the run that produced it. Supplying such a quantity is a substantial part of what this paper contributes.

Self-supervised representation learning. Masked prediction, in which a model restores randomly hidden parts of its input from the remainder, is the representation ingredient we borrow rather than introduce [9, 47]; the code $c_m(t)$ of Eq. 1 comes from exactly such a model, trained without labels on the joint delay embedding. What this paper adds to that literature is a caution about reading causal structure off these models directly. Scoring a channel by how well a masked autoencoder reconstructs it from the rest is the natural foundation-model-style probe of coupling, and at $V=1001$ on homogeneous pools it inverts: reconstruction measures typicality, so the ghost channel, which is causally connected to nothing by construction, outscores genuinely coupled variables (Mechanism 2). Deep ensembling [28], the standard remedy for an unreliable learned score, averages the optimisation variance away and converges to the shared bias rather than removing it (Mechanism 3). Difference-based readouts fail separately and for a different reason, one that is a property of the model’s function class rather than of any implementation [8], and that reason is sharpened by the same embedding theorem the first paragraph invokes (Mechanism 1). The representations transfer; the causal content has to be secured in the readout.

Surrogate-data and validity testing. The ghost channel descends directly from surrogate-data hypothesis testing, in which a statistic computed on the data is compared against the same statistic computed on series constructed to preserve chosen properties (marginal distribution, autocorrelation, spectrum) and to destroy the rest, including the temporal alignment that any real coupling would need [22, 23]. The difference here is placement, not principle. A surrogate test is conventionally an outer loop: compute the statistic, generate a surrogate ensemble, refit, compare, accept or reject. The ghost is instead a column of the data matrix, present before the

encoder is trained and scored by the same code and the same readout as every real variable. Three consequences follow. It costs one additional channel and one additional ridge fit rather than a second pass over the whole analysis, which is what makes it affordable at the widths of Table 1; its score is reported on the same scale as the results it validates, so the artifact level of an analysis is read off directly rather than converted into a p -value; and because it passes through the shared machinery, it registers artifacts of that machinery, which is how the typicality inversion of Mechanism 2 was caught, and on real data, where there is no membership truth to check against, it is the run’s only built-in warning. The ghost also inherits the surrogate’s assumptions: a circular shift is a valid null only for stationary series, which is why differencing is a prerequisite and why an inflated ghost means the scan must be discarded rather than thresholded (Sections 4 and 11).

Application domains. The deployments in Section 8 borrow their data and their standards of correctness from four established literatures. Whole-brain imaging at cellular resolution, in the immobilised worm [10] and by light-sheet microscopy in the larval zebrafish [15, 24, 25], is what produces recordings with 10^2 to 10^5 simultaneously observed variables, and hence the problem this paper addresses. Connectomics supplies the independent anatomical standard against which a label-free ranking can be checked [11, 12], and the *C. elegans* reversal-command literature supplies the circuit-level interpretation of the neurons the statistic surfaces, including the AVA, AVB, AIB and RIM architecture and RIM’s own intrinsic dynamics [13, 14, 38]. In clinical electrophysiology, seizures are increasingly analysed as disorders of network organisation [43] within the wider functional and effective connectivity framework [42], which is the frame in which a drivenness map of scalp channels is interpretable at all. In climate, the teleconnection literature [39, 40] and storm-track dynamics [27] describe the large-scale structure that our sea-level pressure map reproduces blind on reanalysis fields [18]. These literatures supply the yardsticks rather than the competitors: the statistic is checked against what they already establish, not benchmarked with them.

Position. MACE is an estimator in the Granger and transfer-entropy family, built on a self-supervised representation and audited by a surrogate control, so all three of its parts are borrowed and only their composition is new. That composition is not merely assembled but derived: Section 5 shows the resulting statistic has an exact population value, that the surrogate’s null follows from the same embedding argument rather than being assumed, and that the linear cost in V is a consequence of one sufficient statistic serving every readout. What the composition adds is an operating range and a per-run check: variable-level drivenness at widths where, in our measurements, the pairwise and constraint-based estimators are wall-clock infeasible, sample-starved, mediation-prone or undemonstrated, together with a control channel whose score can falsify an individual scan without recourse to ground truth. What it does not add is a new causal quantity, an edge-level graph, detection of sources, or any identification guarantee beyond those the family already carries; the semantics stay drivenness only, top- k , and conditional on the stationarity and function-class caveats stated in Section 11.

3 Failure modes of learned causal readouts

The gap above is not closed by pointing a larger learned model at the problem, and it is worth being precise about why. The obvious learned readouts do not degrade gracefully as V grows; they fail for three specific and separable reasons, and each reason turns out to prescribe part of the remedy. We isolate them here, before proposing anything, so that the statistic of Section 4

can be read as the construction these failures leave standing rather than as one design choice among many.

All experiments in this section use coupled logistic-map networks (with heterogeneous and stochastic variants) where ground truth is known by construction; full protocols are in the released ledger. Every analysis embeds a *ghost channel*, our standing negative control, in the tradition of surrogate-data hypothesis testing [22, 23]. The ghost is built by taking one real variable’s series and rotating it in time by a large random offset: the rotated copy retains every single-channel property of real data—the same distribution of values, the same autocorrelation, the same spectrum—but its moments no longer line up with anyone else’s, so it is, by construction, causally connected to nothing. A valid analysis must therefore score it at zero; any nonzero ghost score measures the analysis’s own artifact level. This is the logic of surrogate-data testing, embedded as a channel and scored by the same machinery as every real variable.

Mechanism 1 (Maturity collapse of difference-based importance). *Difference-based scores (leave-one-covariate-out and relatives) measure a trained model’s loss when one input is removed. When a cause’s imprint is redundant across several observed effects (generic for dynamical systems, since any single effect’s delay embedding reconstructs its driver [2]), the difference collapses to zero precisely when training succeeds: a mature model re-routes around any single removed input. The separating signature, which distinguishes this collapse from overfitting, is the following. On a hub-structured network, checkpointing each model across its own training trajectory shows redundant-parent scores rising, peaking at partial maturity, and collapsing to zero by epoch 50 in the seed mean ($0.007 \rightarrow 0.026$ at epoch 10, then -0.002), while unique-parent scores grow monotonically throughout ($0.02 \rightarrow 0.28$). The divergence of these two curves is the synthetic signature. Two caveats belong with it, since only two seeds were run. The collapse is complete in one seed (hub gain $+0.034$ at epoch 10 falling to -0.025) and absent in the other, where the hub gain is largest at epoch 50; and the two seeds differ tenfold in unique-parent gain (0.05 against 0.51), so the seed mean understates the spread. The curves also do not cross in the mean: unique-parent gain exceeds redundant-parent gain at every checkpoint. What the mechanism rests on is the collapse of the redundant-parent curve, not an intersection. On real neural data (C. elegans, dense connectome redundancy [11, 12]) the collapse endpoint replicates: mature leave-one-out gains are exactly zero: AUROC 0.500 to three decimals in all three worms tested, and no intermediate checkpoint yields informative scores. This is the function-class dependence of importance gaps [8] expressed along the training trajectory: such scores work because of restricted capacity, not despite it. Figure 1 shows the two curves.*

Mechanism 2 (Typicality substitution in masked reconstruction). *With difference-based scores ruled out by redundancy, the natural move is to stop removing inputs and instead ask how well the rest of the system reconstructs a channel outright.*

Scoring a channel by how well a trained masked autoencoder [9] reconstructs it from the rest is a natural foundation-model-style probe of coupling [9, 47]. At scale, on homogeneous pools, it inverts: on a 1,000-channel benchmark (a coupled minority of ~ 100 among ~ 900 autonomous channels), reconstruction ranks periodic autonomous channels highest (class median 0.79, mean 0.76), the ghost at 0.74, and puts both coupled members (median 0.03) and chaotic autonomous channels (median 0.00) at the bottom. The decoder’s loss-optimal completion of a masked slot is the family-typical trajectory, so reconstruction measures typicality, not coupling. Coupled channels, perturbed off the family manifold by their drive, are the least typical. Bottleneck width ($8 \rightarrow 256$) and masked-only losses fail to repair the inversion, because neither changes the optimal

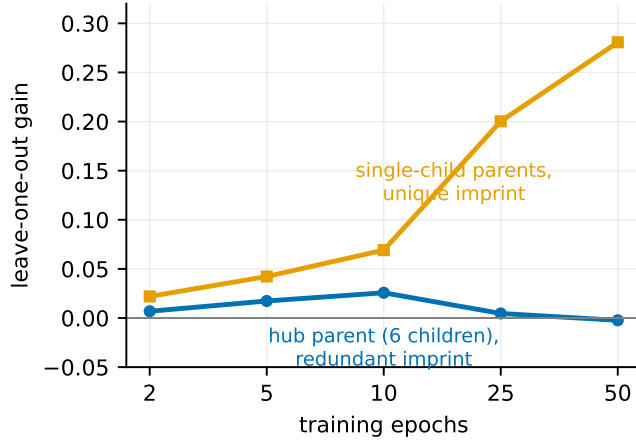


Figure 1: **Maturity collapse is redundancy-ordered, not uniform.** Leave-one-out gain, the increase in a target’s held-out prediction error when one input variable is withheld from an otherwise identical model, plotted against the number of training epochs of the model being probed (mean of two independently generated synthetic networks; epoch values are categorical, not to scale). Gains for the hub parent, whose imprint is redundant across six children, rise, peak at partial maturity, and collapse to zero as the model learns alternative routes; gains for single-child parents, whose imprint is unique, grow monotonically. Uniform degradation—the overfitting signature—is absent.

blind completion (fill-value ablations, run at $V=30$ where no inversion exists, showed no effect there). Figure 2 shows the inversion.

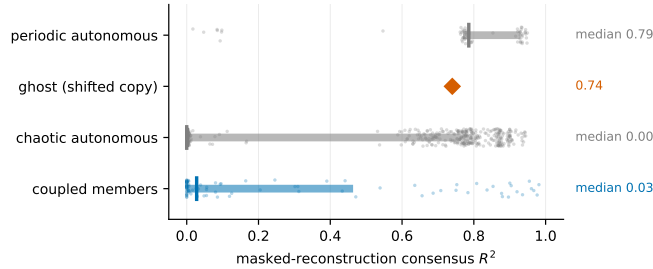


Figure 2: **Masked reconstruction measures typicality, not coupling.** Consensus masked-reconstruction R^2 per channel class at $V=1001$ (8-encoder ensemble). Each point is one channel; the bar spans the interquartile range and the tick marks the median. Periodic autonomous channels (median 0.79) and the ghost (0.74), which carries no causal information by construction, score highest, while genuinely coupled members (0.03) and chaotic autonomous channels (0.00) score lowest. The causal ordering is inverted. The ghost is a single channel and so has one marker rather than a distribution.

Mechanism 3 (Ensembles average variance, not bias). *A natural remedy for a learned readout’s failures is ensembling [28]: training many models from independent initialisations and using their*

consensus. We tested this directly: eight masked autoencoders on one fixed system, only the seed varying. The consensus average-precision curve rises with ensemble size ($0.111 \rightarrow 0.148$) with the expected $1/\sqrt{M}$ deceleration, exactly as the law of large numbers predicts, and converges to a biased limit: the ghost’s reconstruction score is 0.740 ± 0.004 across the eight independent minima. The typicality artifact is a property of data-plus-objective, reproduced in every basin; averaging over minima cannot remove it. Figure 3 shows both curves.

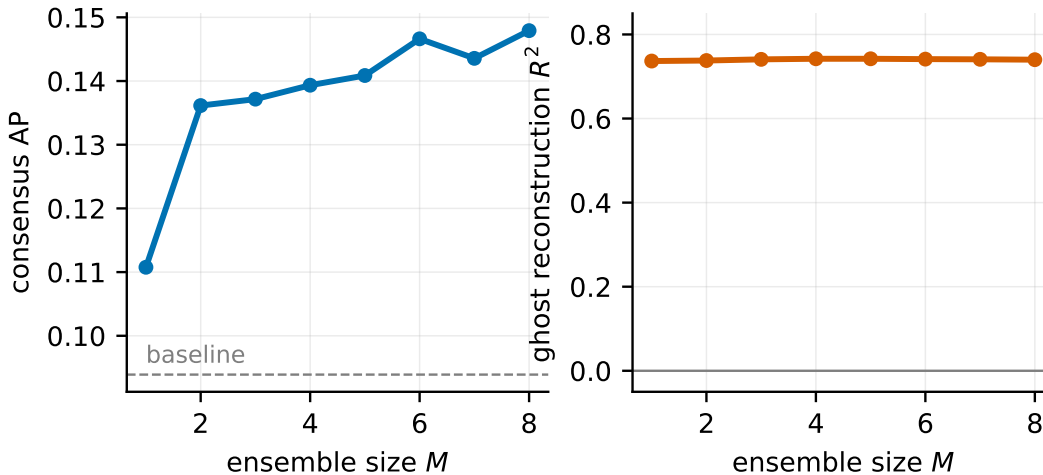


Figure 3: **Ensembling averages variance, not bias.** Left: consensus average precision rises with ensemble size exactly as the law of large numbers predicts, converging toward ~ 0.15 , signal plus shared bias, far below the precision required for detection (dashed line: baseline precision). Right: the ghost channel’s reconstruction score across the same prefix ensembles is 0.740 ± 0.004 : the artifact is reproduced in every independent minimum.

4 Proposed method: MACE

The three mechanisms do not merely motivate the correction below; each rules out a class of readout, and what survives all three rulings is a single construction. Mechanism 1 rules out any score defined as a difference across redundant alternatives, forcing the comparison to be made against a variable’s *own* history rather than against its peers. Mechanism 2 rules out reporting a reconstruction quality absolutely, forcing that own-history term to be *subtracted* rather than merely reported, so that family-typicality cancels instead of ranking. Mechanism 3 rules out the consensus itself as evidence of validity, forcing an artifact meter that is external to the models and re-measured on every dataset—the ghost channel—to be scored by the same machinery as every real variable. A self-referenced, subtractive, externally controlled readout is what remains; the rest of this section specifies it. Let $x_q(t)$ be variable q after per-variable standardisation and first-differencing (stationarity is a prerequisite; on drifting data the circular-shift ghost inflates and the scan must be discarded rather than thresholded, and the ghost must be checked, since it does not announce failure). Form delay embeddings ($E = 3$ throughout) and the joint state $Z(t)$.

We call the complete procedure **MACE**, for *Masked-Autoencoder Conditional Excess*: a masked autoencoder supplies the compressed code, and the statistic it computes is the conditional excess of Eq. 1, the gain in predictability conditional on that code. The ensemble and the embedded ghost control are components of MACE rather than separate procedures.

Code. Figure 4 shows the whole procedure. Train M masked autoencoders (independent initialisations) on Z : random channel subsets are corrupted each batch, the loss is computed on masked positions only [9], and a hard linear bottleneck of width $b \ll \dim Z$ yields codes $c_m(t)$.

Readout. For each variable q and model m ,

$$\text{excess}_m(q) = R^2[x_q(t+1) \mid \phi(\text{own lags of } q) \oplus c_m(t)] - R^2[x_q(t+1) \mid \phi(\text{own lags of } q)], \quad (1)$$

with both terms ridge regressions fit on a training segment and evaluated on a held-out contiguous tail (R^2 clamped at zero), and ϕ a polynomial feature map (degree 2 on the quadratic-map synthetics, where it is exact; degree 3 on all real data; both degrees shown on the heterogeneous pool). The reported score is the consensus mean over models. The subtraction cancels typicality by construction (a channel predictable from its own structure gains nothing from the code); the statistic is not a difference across redundant alternatives, so Mechanism 1 does not apply (the architecture-identical absolute readout was empirically stable across 5–100 encoder-training epochs at $V=30$; the excess readout reuses those saved encoders); and its lineage is explicit: a Granger/transfer-entropy-type quantity [5, 6, 29] — the two are equivalent for Gaussian variables [29] amortised to large V by unsupervised compression [45]—CCM’s convergence-with-library-size criterion makes the analogous self-control implicitly [1]; Eq. 1 makes it explicit and scalable.

Semantics. Positive excess means the system’s state carries information about X ’s next step beyond X ’s own past: X is *driven*. We call this property *drivenness*, a variable’s rank under the statistic its drivenness percentile (0–1 scale), and the top- k variables by consensus excess a system’s *driven core*. A pure source is perfectly predicted by its own history and is therefore invisible. This is a designed, testable asymmetry: known-exogenous variables must score zero (verified below on solar forcing; a genetic silencing approximates it, since the silenced hub falls sharply but not to zero, retaining residual input).

The self-model’s function class is critical. With a linear self-baseline the correction fails (a linear fit of a quadratic map leaves residual self-structure that the code then “explains”, scoring periodic channels +0.26, which is Mechanism 2 returning under a weakened self-model: an under-powered own-history fit leaves exactly the family-typical structure that the code can then re-supply, and the typicality ordering reappears inside the excess itself, periodic channels first); with polynomial features matched to the dynamics (degree 2 suffices on the quadratic maps), self-baselines match theory to three decimals and the confound is cancelled rather than suppressed. On real data, where no exact function class exists, the ghost and known-root channels are the practical check.

Cost, and why it is the practical argument. Every result in this paper was produced on one consumer laptop with a single RTX 3070 GPU (8 GB) and 28 GB of system memory. No

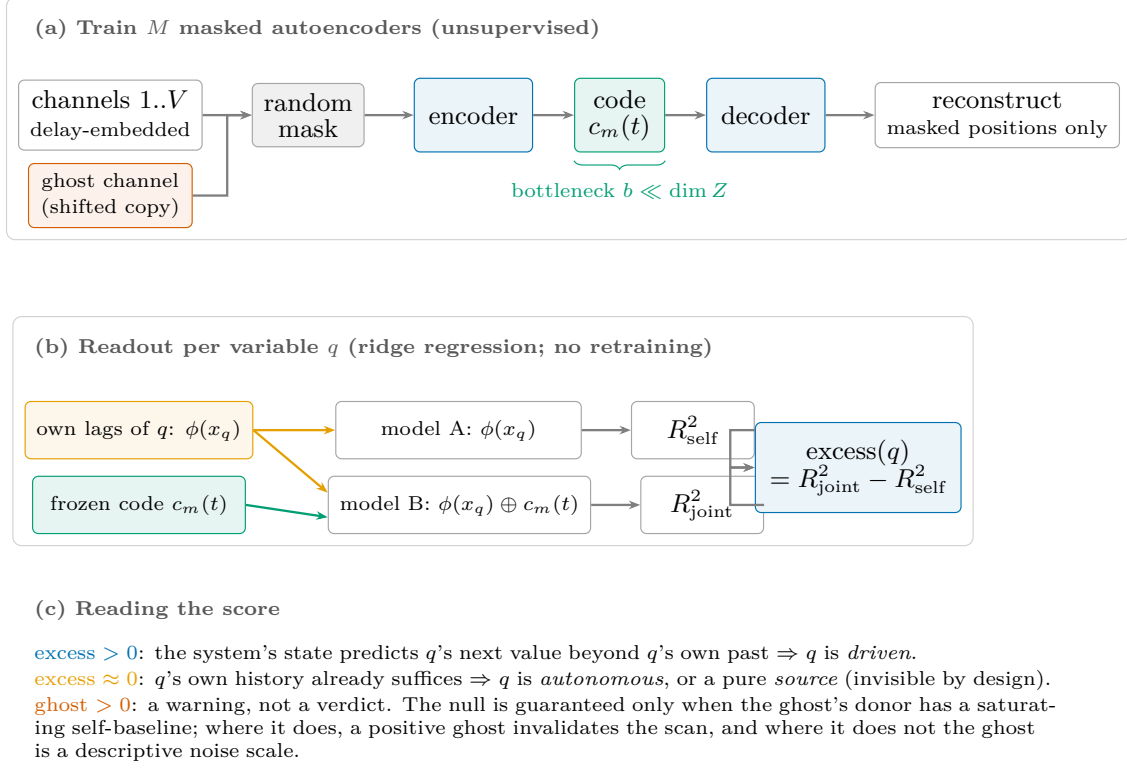


Figure 4: **How MACE is computed.** (a) M masked autoencoders are trained once per system, unsupervised, on the delay-embedded joint state. Random channel subsets are corrupted each batch and the loss is taken on masked positions only, so the hard linear bottleneck of width $b \ll \dim Z$ must retain whatever predicts any channel from the rest. The ghost channel is present as an ordinary column throughout, so it passes through the same encoder, the same code and the same readout as every real variable. (b) Scoring adds no training. For each variable q the frozen code is appended to a polynomial expansion of q 's own lags, and two ridge regressions differ only in whether the code is present; their held-out R^2 values are subtracted. This is why the cost of one more variable is two ridge solves, and why a ghost costs exactly what a variable costs. (c) Reading the result. One qualification to the last line, established in Section 5.5 after this diagram was drawn: a non-zero ghost falsifies the analysis only when the ghost's donor has a saturating self-baseline. A ghost built from a driven donor carries no such guarantee, and on recordings where no channel saturates, as in the worm data here, the ghost is a descriptive noise scale rather than a validity test.

cluster, no multi-GPU node, and no cloud budget was used at any point. Table 1 gives the measured wall-clock costs.

Table 1: **Measured cost of a complete MACE analysis**, on one consumer laptop (RTX 3070, 8 GB). Training times are from archived model timestamps; readout is a ridge fit per variable. The final column is the cost of the pairwise alternative at the same width, from our own wall-clock measurement of CCM (~ 1.1 s per unordered pair, i.e. per call, which returns both directions), costed over the $V(V-1)/2$ calls a non-redundant scan requires.

deployment	V	encoder training	full readout	pairwise equivalent
worm (per animal)	~ 130	< 1 min	seconds	~ 2.7 h
EEG (per window)	24	seconds	seconds	minutes
climate	10,512	~ 5 min	~ 16 min	$\approx 17,000$ h
zebrafish	71,721	~ 4 min	~ 10 min	$\approx 800,000$ h

Three properties produce this. First, the encoder is trained *once* per system and is unsupervised, so its cost is independent of how many variables one subsequently queries. Second, each per-variable readout is a ridge regression on a handful of features, at ~ 8 ms per variable, so scoring scales linearly in V rather than quadratically. Third, saved encoder weights make any new readout (a different self-baseline, a new statistic, a re-analysis under a reviewer’s suggestion) an evaluation rather than a retrain, at seconds per variable-sweep.

Control. The ghost of Section 3 is a component of the statistic, not an auxiliary experiment: one real variable is duplicated, circularly shifted by a large random offset, and passed through the encoder and the readout alongside every other variable, so each scan reports its own artifact floor without reference to ground truth. Ensembling cannot serve this role. Mechanism 3 showed that a data-plus-objective bias is reproduced in every independent minimum, so agreement across models, seeds or bottleneck widths measures only variance; the consensus mean above is therefore used for what it can do—variance reduction, and a cross-model spread diagnostic—while validity is delegated to the ghost. A scan whose ghost does not sit at or below zero is discarded rather than thresholded, with the qualification established in Section 5.5: that rule has force only when the ghost’s donor has a saturating self-baseline, which on the real recordings here it does not.

The consequence is worth stating plainly, because it determines who can use this. A pairwise scan of the zebrafish recording would require roughly ninety GPU-years; the same question, asked through MACE, took fourteen minutes on a laptop that also ran the authors’ text editor. The method is not merely faster than the alternative at scale, it is the difference between an analysis that a small laboratory can run on existing hardware and one that no laboratory would attempt.

5 Why the statistic works

The construction of Section 4 was reached by elimination, and elimination shows only that the alternatives fail, not that what survives is correct. This section supplies the missing account. Under stated assumptions the excess statistic has an exact population value; that value is zero for autonomous variables and strictly positive for driven ones; the ghost’s null follows as a corollary rather than as a hope; and the amortisation that makes the method scale is a consequence of the same embedding theorem that motivates the whole approach. The assumptions are restrictive

enough to be worth stating carefully, because each one that fails corresponds to a failure we have measured.

5.1 Setting and definitions

Let the system evolve deterministically, $s_{t+1} = \Phi(s_t)$, on a compact invariant set $\mathcal{M} \subset \mathbb{R}^m$, observed through smooth scalar observables $x_q(t) = h_q(s_t)$ for $q = 1, \dots, V$. Write $\phi_q(t) = (x_q(t), x_q(t - \tau), \dots, x_q(t - (E - 1)\tau))$ for the delay vector of variable q , and let $c(t)$ denote the code that Section 4 computes from the joint state.

Definition 1 (Autonomy). Variable q is *autonomous* if it is generated by a closed subsystem: there exist a compact invariant set $\mathcal{M}_q$ of box-counting dimension d_q , a smooth invertible map ψ , and a smooth observable η such that $u_{t+1} = \psi(u_t)$ on $\mathcal{M}_q$ and $x_q(t) = \eta(u_t)$. Otherwise q is *driven*. Invertibility is what lets the delay vector, which looks backwards in time, determine the current state; it holds for the sampled flows of the real deployments, whose time- Δ maps are diffeomorphisms. The discrete maps of the synthetic systems are non-invertible endomorphisms, but there the state *is* the observable, so the conclusion holds directly and the hypothesis is not needed.

This is the property the statistic targets, and it is worth distinguishing from the graph-theoretic notion it is often confused with. Autonomy is not “has no incoming edge”; it is “its future is fixed by its own past”. The two differ precisely on sources, which is the asymmetry Section 4 declared by design and which Proposition 1 now derives.

5.2 The self-baseline saturates on autonomous variables

Proposition 1 (Saturation). *Let q be autonomous in the sense of Definition 1 with $E > 2d_q$, and let (η, ψ) be generic. Then there is a continuous F with $x_q(t + 1) = F(\phi_q(t))$ for every t . Consequently, for any function class dense in the continuous functions on $\phi_q(\mathcal{M}_q)$, the self-baseline term of Eq. 1 attains $R^2 = 1$, and for any code $c(t)$ whatsoever*

$$\text{excess}(q) \leq 0, \tag{2}$$

with equality when the class attains its supremum.

Proof. By the embedding theorem in the box-counting form of Sauer, Yorke and Casdagli [3], which extends Takens’ original manifold result [2], for prevalent (η, ψ) and $E > 2d_q$ the delay map ϕ_q restricted to $\mathcal{M}_q$ is an embedding, hence a homeomorphism onto its compact image. Therefore $x_q(t + 1) = \eta(\psi(u_t)) = (\eta \circ \psi \circ \phi_q^{-1})(\phi_q(t))$, a composition of continuous maps, which defines the required F . The self-baseline term then attains $R^2 = 1$. Since $R^2 \leq 1$ for the joint term as well, the difference in Eq. 1 is at most zero. $\square$

Three consequences deserve naming, because each answers a question that the empirical sections could only answer by observation.

First, the complexity confound cancels *exactly*, not approximately. A periodic channel and a chaotic channel both have self-baseline 1, so both have excess 0. The fix that Mechanism 2 motivated is therefore a theorem, which is why autonomous channels in Section 7 pin at zero rather than merely scoring low.

Second, the cancellation holds for any code at all. No property of the autoencoder, its width, its training length or its optimisation quality is needed for the null. This is what makes

the statistic safe to compute with a learned component: the learned part can only create false negatives by being uninformative, never false positives by being wrong.

Third, the null is one-sided. Autonomy bounds excess above by zero and says nothing below it, which is why the detection test in Section 10 is one-sided and why the negative tail of the observed distribution carries no information about drivenness.

5.3 What the statistic estimates

Proposition 2 (Identification). *Suppose (i) the code is sufficient for the state, meaning s_t is measurable with respect to $c(t)$, and (ii) the joint readout class is dense in the continuous functions on the code’s range. Then, with expectations over t ,*

$$\text{excess}(q) = \frac{\mathbb{E}[\text{Var}(x_q(t+1) \mid \phi_q(t))]}{\text{Var}(x_q(t+1))}. \quad (3)$$

Proof. Sufficiency gives that $x_q(t+1) = h_q(\Phi(s_t))$ is a continuous function of $c(t)$; by (ii) the joint readout attains it, so the joint term reaches $R^2 = 1$. The self term equals $1 - \mathbb{E}[\text{Var}(x_q(t+1) \mid \phi_q(t))]/\text{Var}(x_q(t+1))$ by definition of R^2 . Subtracting gives the claim. $\square$

So the identified quantity is **the fraction of a variable’s next-step variance that its own history fails to explain**, certified by the code as explainable by the rest of the system. It lies in $[0, 1]$, it is zero if and only if q is autonomous (Proposition 1), and it is the variance-based analogue of the transfer entropy from the whole system into q , the two coinciding for Gaussian variables [29].

Hypothesis (ii) is worth separating from hypothesis (i), because our own estimator does not satisfy it. Eq. 1 applies the polynomial map ϕ to the own lags only and appends the code *raw*, so the joint readout is affine in $c(t)$, whereas the code enters the true predictor nonlinearly. What MACE computes is therefore a *lower bound* on the quantity Proposition 2 identifies. The size of the gap is measurable, and we measure it. Substituting an oracle code (the full raw system state, so hypothesis (i) holds exactly), abundant samples and a negligible ridge penalty, a driven channel at coupling 0.30 has identified value 0.0443, which the affine-in-code readout recovers as 0.0394 while a readout carrying own-by-code interaction terms recovers 0.0443 to within 10^{-12} . Across the channels and couplings checked, the affine estimator falls 10–16% short even with a perfect code.

The direction of that error matters more than its size, and it is the same direction as every other approximation in this construction: an inadequate joint readout lowers the joint term, hence lowers excess, hence produces false negatives and never false positives. Proposition 1, which supplies the no-false-positive guarantee, needs only $R^2 \leq 1$ for the joint term and is untouched by (ii). Adding interaction features would tighten the bound and is the obvious next refinement; we did not use them here because they multiply the readout’s feature count by the bottleneck width.

This closed form accounts for three patterns that the experiments report and that would otherwise look like quirks.

Drivenness is graded, not binary. The more of a variable’s next step is fixed externally, the larger the conditional variance given its own past. The prediction is directional and its sharpest form holds exactly: no channel of in-degree zero is ever flagged (0 of 81 over the three distinct generating graphs, and 0 of 106 including the heterogeneous re-analysis). Its graded form does *not* hold at the resolution we can measure, and we report that rather than the favourable half:

among channels that are genuinely driven, those the statistic recovers and those it misses are indistinguishable in in-degree (1.60 against 1.66, $n = 63$ and 156 over the three distinct graphs, Mann–Whitney $p = 0.79$ one-sided). Parent count is evidently too coarse a proxy for drive share, which Eq. 1 measures in variance rather than in edges.

The collapse near synchrony is the correct answer, not a failure. As coupling drives a target toward synchrony with its driver, the driver’s signal becomes recoverable from the target’s own past, the conditional variance falls to zero, and excess vanishes. The statistic is answering the question it was asked, and the answer is that own history now suffices.

Sources score zero because their own history determines them. A pure source satisfies Definition 1, so Proposition 1 applies. Measured: zero of 63 sources flagged over three distinct generating graphs, and zero of 82 including the heterogeneous re-analysis.

5.4 Why a single code serves all V readouts

The scaling claim is a corollary rather than an engineering convenience. Every readout needs the same conditioning object, because $x_q(t+1) = h_q(\Phi(s_t))$ holds for all q simultaneously with the same s_t . One sufficient statistic therefore serves all V regressions at once. By the embedding results of Whitney, Takens and Sauer et al. [2, 3], an attractor of box-counting dimension d embeds in $\mathbb{R}^{2d+1}$, so a bottleneck of width $b \geq 2d+1$ suffices *regardless of V* .

That is the whole of the amortisation argument. The encoder is trained once; each additional variable costs one ridge solve in $b + \dim \phi$ features. Pairwise methods instead estimate $O(V^2)$ separate conditional relations, each with its own embedding and its own convergence check, which is the cost Section 1 measures rather than assumes.

The bottleneck is therefore not regularisation by hope. The requirement $b \geq 2d+1$ is a statement about the attractor, and d does not grow with V for systems whose collective dynamics are low-dimensional. Where it does grow, in spatially extended systems with extensive attractor dimension, the same theory predicts that b must grow with it; that is a testable boundary and we have not mapped it.

A finite-sample version of the same argument explains why the raw alternative fails. Conditioning on all VE raw dimensions carries estimation variance growing with VE/n , whereas the code carries b/n . Measured: the raw-ridge arm reaches average precision 0.10 against 0.24 for the coded readout, on identical data and identical regularisation tuning.

5.5 The ghost’s null is a corollary, and it constrains the donor

A ghost is a circularly shifted copy, $x_g(t) = x_D((t + \Delta) \bmod n)$, of a donor channel D . If D is autonomous, the shifted copy satisfies the same recursion and is therefore autonomous itself, so Proposition 1 applies directly and the ghost’s excess is at most zero for any code. **The ghost’s null is a theorem, not a calibration**, and that is what makes it usable as a validity check: a ghost scoring above zero falsifies a premise of the analysis rather than merely looking unusual. The premises available to fail are stationarity, under which the shifted copy no longer satisfies the same recursion; function-class adequacy, under which the self-baseline does not attain its supremum; and split hygiene.

The converse is where our own archived runs were wrong, and the theory predicts the error precisely. If the donor is *driven*, its self-baseline does not saturate, Proposition 1 does not apply, and nothing bounds the ghost’s excess from above. The runs reported in Section 7 drew their single donor uniformly from the coupled web, which is exactly the population for which the

guarantee fails. Scoring a panel of 200 ghosts per system on the archived encoders confirms the prediction: of 800 ghosts across four runs, every one exceeding $+0.001$ had a donor drawn from the coupled web with one-step self-predictability between 0.84 and 0.95, whereas ghosts whose donors were self-predictable above 0.99 never exceeded $+0.0004$.

Donor autonomy is not observable. But the property Proposition 1 actually requires, a saturating self-baseline, is directly measurable with no ground truth at all, because it is the second term of Eq. 1 and is computed anyway. This yields the corrected procedure of Section 10: draw ghosts only from channels whose self-baseline saturates.

5.6 Finite samples: the null distribution and what it licenses

In population the excess of a driven variable is non-negative. In a finite sample the joint regression carries b additional parameters, and on held-out data those cost accuracy whenever they carry no signal, so the sampling null sits slightly below zero with a heavy lower tail. Measured across the same 800 ghosts: median between -2.6×10^{-5} and -2.5×10^{-4} by system, fifth percentile as low as -2.8×10^{-2} , and minimum -0.27 . Stochastic donors dominate that lower tail, exactly as Proposition 1 implies, since an AR(1) channel’s self-baseline never saturates: in the heterogeneous pool the AR(1) donors have mean ghost excess -0.022 against -0.00004 for the logistic donors.

Two things follow. The lower tail must be discarded rather than interpreted: a large negative excess means the code was an expensive irrelevance for that channel, not that the channel is somehow anti-driven. And the ghost panel is a genuine permutation null, each ghost being an independent draw from the sampling distribution of the statistic under autonomy, with the marginal distribution, autocorrelation and spectrum of a real channel preserved and only the temporal alignment destroyed, in the manner of Theiler-type surrogates [22]. Thresholding at the panel’s upper extreme is then a one-sided permutation test at per-channel level approximately $1/(K + 1)$ for K ghosts.

The cost of that panel is the argument’s convenient consequence. A ghost is two ridge solves against an already-computed code, which is precisely the cost of one more variable. A panel of K ghosts therefore costs what K additional channels cost, so a method that scans 7×10^4 variables in minutes can afford a null of thousands. We do not claim family-wise control across V channels; multiplicity still needs handling in the usual way, and we report top- k results and the panel threshold rather than a corrected p -value.

5.7 What the theory does not deliver

Function class. Proposition 1 requires the self-model class to attain the saturating supremum. Where it does not, the code completes what the baseline could not and autonomous channels acquire spurious positive excess. This is not hypothetical: a linear own-lag baseline on quadratic maps gave self- $R^2 = 0.62$ where the theory demands 1, and the typicality confound reappeared in full. Degree-2 polynomials represent the logistic family exactly, which is why they are used on the synthetics, and degree 3 is used on real data where the true class is unknown.

Sufficiency. Proposition 2 assumes the code determines the state. A lossy code makes the joint term fall short of 1, so excess is under-estimated. The error is therefore one-directional: a lossy code produces false negatives, never false positives. This is consistent with the recall measured in Section 7.1 and is the reason we treat the statistic as a detector rather than an estimator of effect size.

Edges. Neither proposition identifies *which* variables drive q . Excess is a statement about the whole system’s information about q , and attributing it to individual parents is a different problem which Mechanism 1 gives reason to expect is hard.

Determinism and stationarity. The Takens step needs a deterministic finite-dimensional attractor. Observational noise makes the embedding approximate and the saturation partial, so on real data an autonomous channel’s excess is near zero rather than exactly zero, and the panel rather than the value zero must set the threshold. Stationarity is assumed throughout, and the ghost is the check that it held.

How partial that saturation becomes on real recordings is not a detail, and it is the largest gap between this section and Section 8. Across all ten *C. elegans* recordings the self-baseline has median R^2 between 0.179 and 0.222 with a maximum of 0.352: **not one of 1,276 neuron-channels reaches $R^2 = 0.9$, let alone the 0.99 that Section 10 asks of a ghost donor.** The premise of Proposition 1 is therefore not weakly met on the worm, it is absent, and with it go the no-false-positive guarantee and the theorem-backed status of the ghost. We state the consequence rather than leave it to be inferred: **on every real dataset in this paper the ghost is an empirical artifact meter rather than a corollary, and the real-data findings rest on their pre-registered gates and their external anatomical, interventional and physical checks alone.** The synthetic systems, where the premise does hold, are what license the theory; the real deployments are evidence that the statistic travels, not that the guarantee travels with it. Whether a delay embedding of calcium fluorescence can saturate at all at these sampling rates and noise levels is a question this paper does not settle, and it is the first thing we would want measured next.

6 Data and methods

Everything below is fixed in the released scripts; this section collects it in one place so that a reader can reproduce or contest any number without reading the code. Table 2 gives the datasets and their preprocessing and Table 3 every setting that varies between them. Values that differ between deployments are given per deployment, because two of them (the embedding delay and the bottleneck width) are load-bearing in the arguments of Section 5.

6.1 Datasets

6.2 Hyperparameters

Encoder. A masked autoencoder with a hard linear bottleneck: a tanh encoder stack onto a linear bottleneck of width b , a tanh decoder, trained with mean squared error on masked positions only, mask fraction 0.25, Adam for 25 epochs with the learning rate and batch size of Table 3, and M independent initialisations per deployment. The encoder’s hidden width is $\max(2b, 64)$, giving 256 on the synthetics and 128 on climate. Weights are archived so that every readout in this paper is recomputable without retraining.

Readout. Both terms of Eq. 1 are ridge regressions with $\alpha = 1$, fit on the training segment and evaluated on the held-out tail, R^2 clamped at zero before differencing. The self-baseline uses polynomial features of the own lags at the degree in Table 3; the code is appended raw, which is the restriction Section 5.3 analyses.

Table 2: **Datasets, all publicly available without registration.** n is the length of the analysed series after preprocessing. Every *real* series is per-variable standardised on training statistics only and first-differenced; the synthetic maps are standardised but not differenced, being stationary by construction on a bounded invariant set. Stationarity, not differencing, is what the circular-shift ghost requires.

dataset	V	n	source and preprocessing
synthetic networks	1,001	2,000	coupled logistic maps on a sparse random DAG, 3 generating graphs plus one heterogeneous-background re-analysis; membership known by construction
<i>C. elegans</i>	109–139	$\sim 3,100$	whole-brain calcium imaging, immobilised, ~ 2.9 Hz [10]; 5 wild-type and 5 AVA:HisCl animals; $\Delta F/F$, differenced
zebrafish	71,721	2,241	light-sheet whole-brain calcium imaging [15, 24, 25]; differenced
EEG (chb01)	23	$\geq 2,000$	scalp EEG at 256 Hz [16], mean-pooled 4 $\times$ to 64 Hz; 14 windows of 31–101 s spanning annotated seizures and matched interictal spans; differenced
climate	10,512	$\sim 28,000$	NCEP/NCAR reanalysis daily sea-level pressure, 77 years [18]; deseasonalised and differenced

Table 3: **Every setting that varies across deployments.** E is the embedding dimension, τ the delay in samples, b the bottleneck width, M the ensemble size, “degree” the order of the polynomial self-baseline ϕ , and lr and batch the optimiser settings, which are *not* common to all five. Encoder architecture and splits are shared and given in the text. None of these were tuned against the reported outcome: each was declared in the script header before the run.

deployment	E	τ	b	M	epochs	degree	lr	batch
synthetic ($V=1001$)	3	1	128	8	25	2 and 3	3×10^{-3}	64
<i>C. elegans</i>	3	3	32	8	25	3	3×10^{-3}	64
zebrafish	3	1	64	4	25	3	1×10^{-3}	16
EEG	3	1	8	4	25	3	3×10^{-3}	64
climate	3	1	64	4	25	3	1×10^{-3}	128

Splits. Contiguous three-way splits in time, 0.6/0.2/0.2, with an embedding-span embargo of E samples at each seam so that no delay vector straddles a boundary. Standardisation statistics are computed on the training segment only, and the same split governs both the encoder and the readouts.

Delay selection. τ was fixed per dataset on physical grounds and declared before analysis rather than selected against any outcome: $\tau = 3$ for calcium imaging at ~ 2.9 Hz, which is ~ 1 s and the timescale of the dynamics of interest, and $\tau = 1$ elsewhere. $E = 3$ throughout. We did not run a data-driven selection rule such as the first minimum of the auto-mutual-information; a reader applying the method to a new system may prefer to, and Section 10 states the diagnostic that decides whether the resulting embedding is adequate.

Hardware. One consumer laptop with a single RTX 3070 (8 GB) and 28 GB of system memory, running TensorFlow/Keras and scikit-learn. No cluster, multi-GPU node or cloud budget was used at any point in this work.

7 Validation on systems with known ground truth

Whether those three constraints were actually met is not something the construction can settle for itself; it needs systems whose driven set is fixed independently of the statistic, and a stated way to fail.

We validate on systems of $V \in \{30, 1000\}$ logistic-map channels in which a minority sub-web is coupled (sparse random DAG) and the remaining channels are autonomous; membership is known by construction. At $V=1001$ (93–95 members by seed; baseline precision ≈ 0.094):

Table 4: **Top- k precision and false alarms at $V=1001$.** A false alarm is an autonomous channel scoring above the ghost’s *magnitude*, $|\text{excess}(g)|$; the signed ghost value is not a threshold, since it is negative and sits at the 4th–22nd percentile of all scores, so almost every channel exceeds it. The heterogeneous pool is reported at both polynomial degrees because the degree-2 readout does not represent the sine family exactly, and that failure is visible here rather than hidden.

system	degree	precision@10	precision@20	false alarms
seed 0	2	10/10	16/20	0 / 900
seed 1	2	10/10	20/20	0 / 900
seed 2	2	10/10	20/20	0 / 900
heterogeneous pool	2	10/10	18/20	43 / 900
heterogeneous pool	3	10/10	18/20	0 / 900

The false-alarm column is where the degree-2 readout’s limits show. On the three homogeneous systems no autonomous channel clears the ghost magnitude at all, and the largest autonomous score is $+7.3 \times 10^{-6}$ against a weakest top-20 member of $+2.1 \times 10^{-5}$. On the heterogeneous pool at degree 2, however, 43 autonomous channels clear it, *all of them from the sine family*, and the largest autonomous score ($+4.0 \times 10^{-4}$) slightly *exceeds* that system’s weakest top-20 member ($+3.7 \times 10^{-4}$): two sine channels hold ranks 18–19, which is exactly why precision@20 is

18/20 rather than 20/20. Raising the readout to degree 3 on the same data removes every one of these (43 to 0) and restores a $15\times$ margin. This is the function-class dependence of Section 5.7 appearing precisely where the theory says it should, on the one family the degree-2 basis cannot represent, and it is the reason degree 3 is used on all real data (Figure 5).

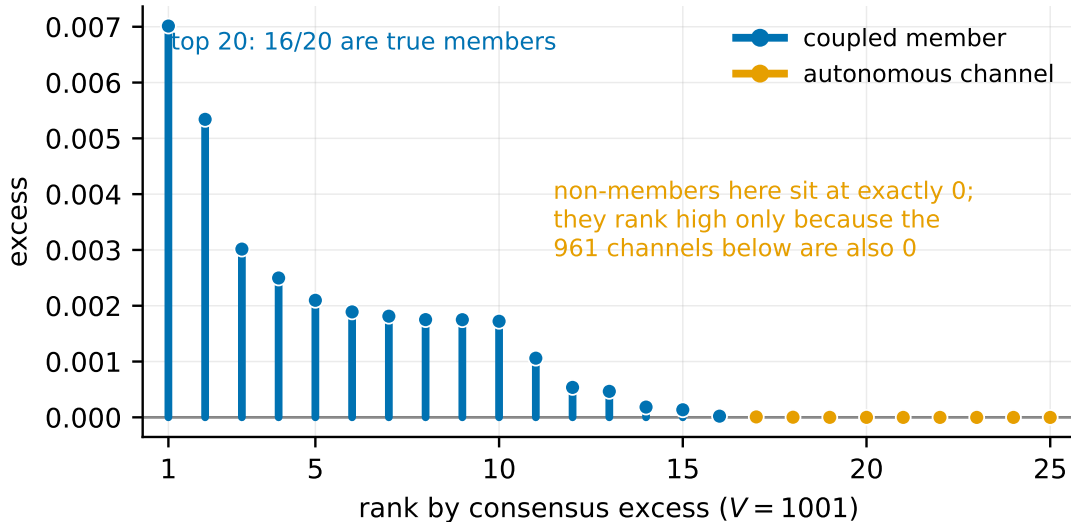


Figure 5: **The excess statistic at $V=1001$ (system seed 0).** The 25 highest-ranked channels by consensus excess, each drawn as a stem with a marker at its value so that channels scoring exactly zero remain visible. Ranks 1–16 are all coupled members (blue). The autonomous channels that appear inside the top 25 (orange) sit at exactly zero: they are ranked high only because the 961 channels below them are also zero, not because they earned a positive score. This is what makes the statistic a top- k detector rather than a ranker: weakly driven members also fall to zero and are therefore indistinguishable from non-members, which is why full-population AUROC is 0.28 while precision within the top 10 is perfect.

Two boundaries of the statistic were measured directly. First, it is a **top- k detector** (members whose drive share falls below the readout’s noise floor receive slightly negative excess and rank below the zero-pinned non-members, so full-population AUROC is uninformative even when top- k precision is perfect—0.28 here by mechanism, not accident; average precision and top- k against stated baselines are the correct metrics). Second, the **encoder is necessary** (the same readout conditioning on all raw dimensions with tuned regularisation fails, AP 0.10 vs. 0.24). Third, **ensembling is helpful but not critical** (archived single-model AP 0.25 vs. consensus 0.24; its chief value is the cross-model variance diagnostic). “False alarm” throughout means an autonomous channel whose excess exceeds the ghost’s magnitude, $|\text{excess}(g)|$, which is the rule Table 4 applies and the one a user without ground truth can apply. The largest autonomous excess observed in any synthetic system is $+4.0 \times 10^{-4}$, in the degree-2 heterogeneous pool, and it is not below the weakest top-20 member of that system but marginally above it; at degree 3 the largest is $+1.3 \times 10^{-5}$, comfortably below.

7.1 Recall, and what a null result does not mean

Top- k precision says what a positive is worth. It says nothing about what a silence is worth, which is the question a user faces whenever a variable of interest fails to appear in the driven core. We measure that here, on the same systems, because it is the number a reader needs before acting on an absence.

The accounting requires one correction first. The membership label used above marks a channel as positive when it has any edge in the generating graph, in or out. A pure source has out-edges and no in-edges, so it is labelled positive while receiving no drive at all, and Proposition 1 requires the statistic to be silent on it. Charging the method for that silence would be charging it for obeying its own semantics, so we report recall against the *driven* subset and report the other roles separately.

A second correction concerns how many systems there are. The heterogeneous pool is built on system seed 0, and its coupled web is not merely similar to that of the homogeneous seed-0 system but *identical*: the two builders draw the same generating graph and simulate it with the same seed, so the first 100 channels agree exactly (maximum absolute difference 0), and only the autonomous background differs. Pooling the two would therefore count the same 100 channels twice. We report the three distinct generating graphs as the primary result and the heterogeneous pool separately, as what it is: a re-analysis of one of those graphs against a harder autonomous background.

Using the panel threshold of Section 10, over the three distinct generating graphs, Table 5 reports recall by the role each channel plays in the generating graph:

Table 5: **Recall by a channel’s role in the generating graph**, over the three distinct generating graphs, at the panel threshold of Section 10. Sources and isolated channels are required to be silent by Proposition 1; they are reported separately from recall rather than counted against it.

role in generating graph	channels	flagged	recall
driven (has at least one parent)	219	63	29%
source (parents none, children some)	63	0	0% (required)
isolated (no edges)	18	0	0% (required)

At that operating point precision is 1.00 on these three systems (63 true positives, no false positives). The heterogeneous re-analysis of seed 0 is the harder case and the only one that produces false alarms: 18 true positives against 4 false positives, precision 0.82, recall 24% of its driven set. The four false alarms are all sine-family autonomous channels, which is the degree-2 function-class failure Section 5.7 predicts: at degree 3 on the same data no autonomous channel exceeds the ghost magnitude at all (43 channels do at degree 2, none at degree 3), and the margin between the largest autonomous score and the weakest top-20 member widens from $0.9\times$ to $15\times$. Taken together the two settings give 81 true positives against 4 false positives, precision 0.95, which is the figure quoted in the abstract; readers who prefer the non-overlapping count should use the 1.00 and 0.82 figures separately. Two readings follow, and they pull in opposite directions.

The favourable one: the statistic never fires on a channel it is theoretically required to miss. Zero of 63 sources and zero of 18 isolated channels were flagged over the three distinct graphs (zero of 82 and zero of 24 if the heterogeneous re-analysis is included), which is the sharpest

available confirmation that the quantity being measured is drivenness rather than participation, coupling strength, or complexity.

The unfavourable one, and the more important for a user: **recall among genuinely driven channels is roughly one in four**. The method finds the most heavily driven quarter and is silent on the rest. Proposition 2 says why this is expected rather than accidental. Excess equals the share of a variable’s next-step variance that its own history cannot explain, so a channel whose drive is weak, or whose drive is largely recoverable from its own past, has a small population value that finite samples cannot separate from zero. We are not able to say which driven channels those are: detected and undetected driven channels are indistinguishable in in-degree (1.60 against 1.66, $n = 63$ and 156 , Mann–Whitney $p = 0.79$), so parent count does not predict recovery and we have no validated rule for anticipating a miss. That is a real gap in the account, and it is why the recommendation below is to treat a null as uninformative rather than to model which nulls are trustworthy.

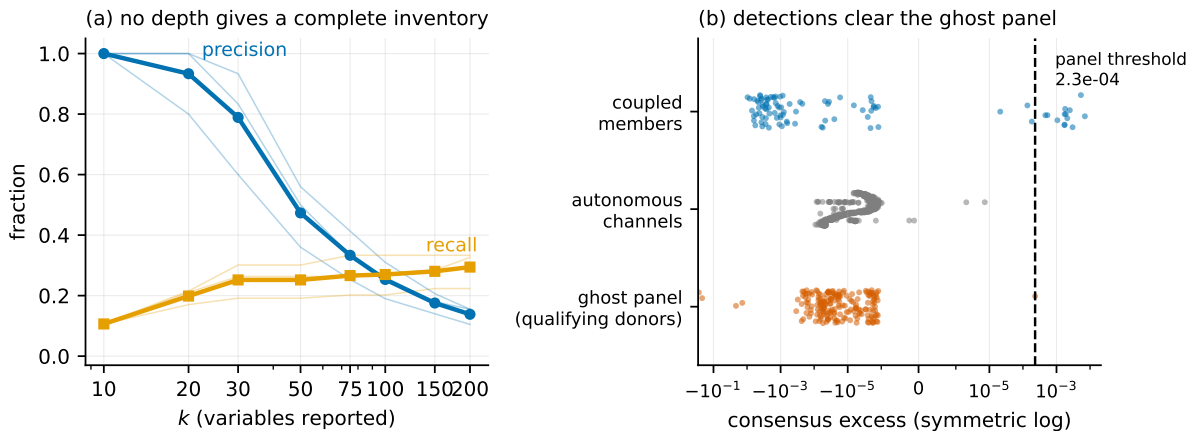


Figure 6: **The cost of the statistic’s precision is recall.** (a) Precision and recall against the number of variables reported, averaged over the three homogeneous systems (faint lines: individual systems). Precision is perfect at $k=10$ and falls steadily, while recall never exceeds 0.30: there is no depth at which the output becomes a complete inventory of the driven set. (b) The same system (seed 0) as a distribution. The ghost panel (orange, 188 qualifying donors) and the 900 autonomous channels (grey) both lie entirely below the panel threshold (dashed), while a clearly separated group of coupled members (blue) clears it. The bulk of members sits inside the null, which is the population of false negatives that (a) quantifies; the negative tail spans four decades and carries no information about drivenness (Section 5.6).

The operating characteristic makes the trade explicit (Figure 6). Averaged over the three homogeneous systems, precision falls from 1.00 at $k=10$ to 0.93 at $k=20$, 0.79 at $k=30$ and 0.25 at $k=100$, while recall against the label rises only from 0.11 to 0.27 across the same range. There is no depth at which this statistic becomes a complete inventory of the driven set. It is a high-precision, low-recall instrument, and the practical consequence is worth stating without hedging: **a variable’s absence from the driven core is not evidence that it is autonomous**. Positive findings from MACE are strong; negative findings are weak, and should be treated as “not detected at this sample size” rather than as “not driven”.

8 Real-world validation and deployments

We deploy the statistic on four open real-world domains.

They are ordered by how much independent ground truth each affords, from most to least, and each therefore tests something its predecessor could not. The worm has named cells, a published connectome and a targeted genetic silencing; the zebrafish has no labels at all, only three-dimensional coordinates, so its only available check is geometric, at $500\times$ the width; clinical EEG has an event annotation but no channel-level truth, and asks whether the statistic tracks a change of regime within one subject; the atmosphere has no truth of any kind, only a forcing that is exogenous by physics and a century of established circulation structure to fall against. All datasets are public and unrestricted. Registration status is stated per study below; the released protocol document and analysis-script headers are the citable artifacts. Table 6 summarises the validity gates across all deployments.

Table 6: **Validity gates across deployments.** The ghost is a circularly shifted real channel (must score ≈ 0); exogenous-root channels are known non-driven variables (must score ≈ 0).

deployment	V	n	ghost	exogenous root
synthetic ($\times 3$ seeds + het.)	1,001	2,000	≤ 0	AR(1) family ≈ 0
worm, wild type ($\times 5$)	109–135	$\sim 3,100$	-0.041 to $+0.004$	—
worm, AVA silenced ($\times 5$)	120–138	$\sim 3,000$	-0.02 to -0.01	AVA falls to percentile 0.38
zebrafish	71,721	2,241	-0.031	—
EEG (chb01, 14 windows)	23	$\geq 2,000$	-0.034 to $+0.022$ (5 of 14 positive)	—
climate	10,512	$\sim 28,000$	-0.005	sunspots -0.003

8.1 *C. elegans*: label-free recovery of the motor-command circuit, and an interventional test

We start where the statistic can be checked hardest: the only system in this paper that offers named cells, an independently mapped wiring diagram, and a perturbation that can be aimed at a specific hub.

Wild type. Whole-brain calcium imaging of five immobilised worms (109–135 neurons, $\sim 3,100$ samples) [10]. Predictions fixed in the analysis script before execution: ghost ≈ 0 ; sensory neurons (hand-curated canonical classes; $n=8/6/1$ per development worm, so per-worm evidence is weak though the direction is uniform) less driven than command/motor neurons in the constant environment; positive correlation with anatomical in-weight [11, 12]. All three passed on the three animals used during analysis development (hereafter *development worms*) and again on two animals reserved since the study’s declaration and opened only after predictions were fixed, with the strongest anatomy correlations of the five ($+0.30$, $+0.34$). The top-10 most-driven neurons are 10/10 command/motor class in two of the three development worms, with the command core topping the two held-out rankings as well: AVA, AVE, RIM, AIB, RIB, RIV, RME, SMDV and ventral-cord motor neurons. This is the canonical motor-command ensemble of the immobilised worm [10], obtained without access to anatomical or functional annotations. The ensemble itself is not a discovery here and we do not present it as one: it is the command

core [10] report from unsupervised decomposition of these same recordings, and it substantially overlaps the hub set that Uzel et al. [32] identified independently and then confirmed causally by silencing. Recovering it is a validation that the statistic finds known structure without labels, not a new claim about the worm.

Intervention. The same pipeline on five worms with the reverse-command hub AVA silenced via a histamine-gated channel [10]. Directions stated in the working ledger before the silenced-worm data was opened: the silenced hub shows the largest decrease within the command core (drivenness percentile $0.92 \rightarrow 0.38$); drivenness declines across the command cycle (AVE -0.28 , RIB -0.29 , VB01 -0.64 , RMED -0.49 , VA01 -0.45). We had first described these as AVA’s module and motor pool, which is anatomically wrong and we correct it: only AVE, which is gap-junction coupled to AVA, and the A-class motor neuron VA01 are AVA’s own targets. VB01 is a B-class motor neuron driven by AVB through gap junctions [37] and RIB is a forward-command neuron whose activity is anticorrelated with AVA [10], so both belong to the *forward* circuit, and RMED is a head motor neuron rather than a documented AVA target. The decline therefore spans both phases of the command cycle, which is what [10] describe as network-wide uncoupling under this manipulation rather than a selective loss of AVA’s own outputs; ghost scores in the silenced cohort lie between -0.018 and -0.008 , i.e. at or below zero in all five animals. Two further observations emerge from the pooled per-cell contrast, and both deserve unpacking rather than a bare number.

RIM and AIB retain drive almost unchanged (-0.08 , -0.08 , class-pooled across the five-animal cohorts, though not uniformly in every single cell, since one silenced animal’s RIMR sits low while its RIML tops the ranking) while every other module member collapses. **This is a recovery of a known dissociation, not a discovery, and it is the more useful for that.** The AVA-silenced recordings are not a new experiment: they are the perturbation published with this dataset [10]. Its authors report that AVA silencing attenuates AVA substantially and uncouples it from the global cycle, attenuates AVE and RIM *slightly* and the A-class motor neurons significantly, but leaves phase relationships with most other neurons normal and the cyclical dynamics and recruitment patterns largely preserved, concluding that AVA “is not a privileged generator of motor commands”. That distinction between attenuated amplitude and preserved coupling is exactly the one our statistic is built to make, since excess measures predictability rather than activity, and it is why RIM can be slightly attenuated in power and still retain drivenness. Independently, Gordus et al. [31] silenced AVA with the same histamine-gated channel and found that AIB and RIM not only retain their responses but respond faster and more reliably without AVA. So the literature already says this pair is not slaved to AVA, and what our result adds is that an unsupervised predictability score, given no labels and no knowledge of the manipulation, reproduces the dissociation. That is a validation of the statistic against an established result, and we present it as such.

The mechanism is better established than our first account of it, and in one respect we had the sign wrong. Piggott et al. [38] show that AIB *inhibits* RIM, and that it is RIM *suppression* that triggers reversals “independently of the AVA/AVD/AVE-mediated stimulatory circuit”, so the AVA-independent route is disinhibitory rather than a driving input as we first wrote. RIM’s two states are real but are not established as intrinsic: [14] attribute the depolarised and hyperpolarised modes to circuit-level mechanisms, glutamate and tyramine signalling in one and gap junctions in the other, and [31] attribute the bimodality of AIB, RIM and AVA jointly to network feedback. We accordingly claim only that RIM and AIB carry drive that does not route through AVA, which the interventional literature supports, and not that RIM is intrinsically

autonomous, which it does not. The identical pattern would also arise from a readout that is simply insensitive to whatever we manipulate, and five animals under one perturbation cannot distinguish the two. Confirming the first reading over the second would need independent replication, a manipulation that targets AIB or RIM directly rather than routing through AVA, and single-cell (not class-pooled) consistency, which the RIML/RIMR split above already shows is not yet in hand.

A candidate reorganisation onto the forward hub AVB (+0.34, which does not clear any null below and sits inside the wild-type-only noise range, so what follows is a mechanism we could not support and retain only to record what was predicted; AVB is recorded in three of the five silenced animals and the rise is carried by two of them, so this observation is the weaker of the pair; ALA moves by a comparable margin, +0.30, Table 7, but its raw excess is negative in both cohorts (−0.031 and −0.016), so that movement lies entirely inside the tail this paper elsewhere declines to interpret; it is reported for completeness and carries no claim). AVB is the forward-command counterpart to AVA within the same architecture, driving forward locomotion as AVA drives reversal, and the same whole-brain recordings that motivated the wild-type analysis established that this worm’s global dynamics organise into a small number of alternating command states rather than one fixed hierarchy [10]. A rise in AVB’s own drivenness once the reversal hub is silenced is the pattern that organisation would predict if the rest of the network reorganised around whichever command pathway remains. We set that reading against the literature rather than arguing around it, and it does not survive the comparison.

Every direct measurement of AVB or of forward locomotion after AVA is removed runs the other way. Meng et al. [36] report that inactivating AVA *reduces* AVB calcium dynamics, and that the two are not strictly reciprocally inhibitory: AVA maintains a tonic extrasynaptic excitation of AVB over exactly the sustained timescale a histamine silencing occupies, so their model predicts AVB should fall. Behaviourally, silencing AVA with this same channel slows forward locomotion by about half [34], and AVA ablation shortens forward runs rather than lengthening them [33]. Anatomy offers no direct route either: AVA receives synaptic input from AVB but makes no direct synapse onto it [37]. Two coupling results point the same way, which matters more for us than the activity results because drivenness is a coupling measure: Meng et al. [36] find that blocking AVA transmission *weakens* the AVA–AVB relationship, and Kato et al. [10] report that AVA silencing uncouples AVA from the global cycle while other neurons’ phase relationships stay normal.

Two things keep this from being a flat contradiction, and neither is sufficient. First, reciprocal inhibition is contested rather than settled: Roberts et al. [33] model it explicitly and fit behaviour well, and take as their motivating puzzle the same paradox at issue here, that silencing reverse-command neurons *reduces* forward dwell time, so the canonical model already predicts non-monotonic responses to removing AVA. Second, no published work measures AVB’s *predictability* under AVA silencing, and one study points our way: Brennan and Proekt [35] find that removing AVA computationally leaves the whole-brain predictive structure intact, and identify AVB as one of only two locomotion neurons with no significant across-animal variability, which is close to saying AVB is the most network-determined of them. A master–slave reading in which AVA is AVB’s dominant phasic driver, whose removal leaves AVB a passive integrator of slower distributed signals, would give lower amplitude and higher predictability at once. That is a hypothesis, nobody has measured it, and with our own signal carried by two of three recorded animals and failing every null in Table 7, it is not one this paper can support. We record the observation and the contrary evidence together, and claim neither.

These per-cell readings need a null, and supplying one changes what can be claimed. Percentiles are within-animal ranks, so a cohort-wide depression of the identified neurons forces every initially high cell downward whether or not it is individually affected; the silenced cohort’s identified neurons do fall as a block, by -0.14 on average. We therefore test each cell against three nulls (Table 7): a within-animal permutation that destroys cell identity while preserving that block shift, asking whether a named cell moved more than an arbitrary one; a permutation over all 252 five-versus-five relabellings of the ten animals, asking whether the wild-type versus silenced split matters at all; and a wild-type-only split-half contrast that measures how large a difference arises between animal groups that differ by nothing.

The intervention effect is real at cohort level and much weaker at cell level than our first reading of it. Three cells clear the cohort-label null uncorrected, VB01 ($p = 0.032$), AVA ($p = 0.008$) and AVE ($p = 0.024$); after a max-statistic correction across the ten contrasted cells *only VB01 survives* ($p = 0.032$), with AVA at $p = 0.135$. The largest single faller is VB01, a motor neuron, not the silenced hub itself. Three of the reported deltas, RIB, AIB and AVB, lie inside the range produced by splitting wild-type animals against each other, so they are not distinguishable from sampling noise. In particular **the AVB rise does not survive**: its $+0.34$ is smaller than the ± 0.34 that wild-type-only splits already produce, so the reorganisation reading above must be treated as a conjecture with no statistical support here, and the RIB and AIB retention is likewise consistent with noise rather than established by it.

What the intervention therefore demonstrates is that a statistic computed with no access to the perturbation registers it at the level of the affected *module*, in the pre-registered direction, with AVA among the two most extreme cells. It does not demonstrate resolution at the level of individual named neurons, and we withdraw that framing. Figure 7 gives the per-cell contrast; the null distributions are released alongside the code.

8.2 Zebrafish: a whole-vertebrate-brain map

The worm tests the statistic against labels. The zebrafish removes them and asks whether the method survives a five-hundred-fold increase in width, where the only check left is whether the driven core is anywhere in particular.

ZAPBench provides $71,721$ segmented neurons $\times$ $7,879$ timesteps of light-sheet imaging in a larval zebrafish [15]. On a 2,241-step window (open-loop, rotation, dark), the full-brain excess map is computed in minutes on a laptop GPU. The ghost scores -0.031 ; $\sim 2,800$ neurons clear the $+0.01$ excess threshold; and the driven core is spatially coherent: the top-1,000 neurons are $1.78\times$ more tightly packed than random draws (median nearest-neighbour distance 16.8 vs. 29.8 ± 0.6), concentrated posteriorly and against the midline, consistent with the hindbrain/midline territory where internally generated motor dynamics are reported to reside [24], in a window dominated by the dark condition. Region names await atlas registration and are not claimed (Figure 8).

8.3 Clinical EEG: a confirmed effect and an instructive null

Both animal deployments had a stable organisation to recover. Clinical recordings do not, and they are where the statistic’s semantics can be violated by the clinical question itself: the quantity clinicians want is the onset zone, which is a source, and sources are invisible here by construction. What remains testable is whether drive reorganises between conditions—and, as it turns out, where that reorganisation stops being resolvable at the level of a single event.

CHB-MIT scalp EEG, subject chb01: 23 channels, 256 Hz; six of the seven annotated seizures

Table 7: **Drivenness percentile, wild type (WT) versus AVA-silenced**, with per-cell nulls (AVA:HisCl, histamine-gated chloride channel; cells identified in ≥ 3 worms of both five-animal cohorts; L/R-collapsed; percentiles on a 0–1 scale; rows ordered by Δ). Raw excess is shown beside the percentile because a percentile is a within-animal rank and can move while the underlying excess stays negative, i.e. inside the tail that Section 5.6 says carries no information about drivenness. Two rows are in that tail: ALA is negative in *both* cohorts, so its +0.30 is movement within the discarded region and not a rise in drivenness, and AVB starts negative in wild type. p_{cohort} permutes the cohort labels over all 252 five-versus-five relabellings of the ten animals with cell identity preserved, and p_{corr} applies a max-statistic correction across the ten cells tested. A third null, permuting percentiles within each animal to destroy cell identity while preserving the cohort-wide shift of -0.14 , is reported in the released results. Only VB01 survives correction; AVA is the second most extreme cell but reaches $p = 0.135$ corrected. The deltas for RIB, AIB and AVB lie inside the range that wild-type-only split-half contrasts already produce, so they are not separable from sampling noise.

cell	percentile			raw excess		p	
	WT	HisCl	Δ	WT	HisCl	cohort	corr.
VB01 (motor)	0.91	0.28	−0.64	+0.038	−0.063	0.032	0.032
AVA (silenced hub)	0.92	0.38	−0.54	+0.059	−0.018	0.008	0.135
RMED (motor)	0.90	0.41	−0.49	+0.037	−0.020	0.119	0.373
VA01 (motor)	0.82	0.37	−0.45	+0.078	−0.017	0.135	0.460
RIB	0.66	0.37	−0.29	+0.023	−0.039	0.302	0.810
AVE	0.97	0.69	−0.28	+0.110	+0.004	0.024	0.825
RIM	0.93	0.84	−0.08	+0.074	+0.034	0.492	1.000
AIB	0.75	0.65	−0.09	+0.040	−0.007	0.659	1.000
ALA	0.16	0.46	+0.30	−0.031	−0.016	0.079	0.794
AVB (forward hub)	0.49	0.82	+0.34	−0.012	+0.007	0.087	0.690

are analysed (the seventh record was not in our retrieved subset) [16, 17], in a field where seizures are increasingly understood as network-organisation disorders [41–43]. One caveat governs all outputs: seizure-onset zones are *sources*, and sources are invisible to this statistic by design. The statistic maps the spread network, not the origin. Results: ghosts across the 14 windows span -0.034 to $+0.022$, and **five of the fourteen are positive** (no numeric tolerance was pre-specified; magnitudes elsewhere range from ~ 0.000 on the synthetics to 0.041 on one wild-type worm). By the sign rule of Section 5.5 those five windows are not clean, and we report them rather than the magnitude summary we first gave, which concealed the sign because its extreme, -0.034 , is the harmless end. Only records 03, 15 and 26 have both windows ghost-clean; restricted to those three the concentration effect below does not reach significance ($p = 0.25$). We therefore report the six-record result as the primary analysis and this restriction beside it, rather than choosing whichever is more favourable; drivenness *concentration* (top-4 channel share) is higher during seizures than between them (0.68 vs. 0.48 over the six matched records; five of six within-record pairs, with the exception, record 03, in the opposite direction; paired one-sided Wilcoxon $p = 0.031$).

That measurement holds, but the interpretation we first gave it does not, and the correction is instructive enough to report in full. The statistic is a share of clamped-positive excess, so both its numerator and its denominator are built from the channels that carry detectable drive.

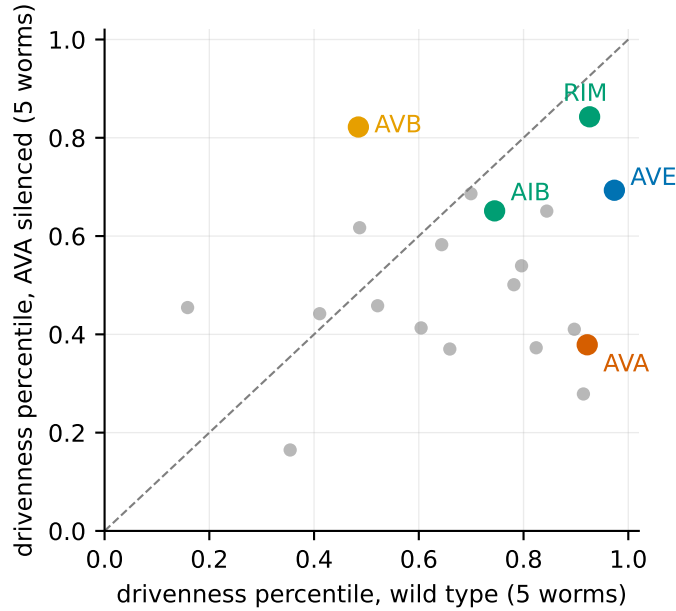


Figure 7: **An observational statistic tracks a genetic intervention.** Per-cell drivenness percentile in wild type (horizontal) versus AVA-silenced animals (vertical); each point is a cell identified in at least three worms of both cohorts. The silenced hub AVA and its module fall below the diagonal; RIM and AIB barely move; the forward hub AVB rises (the AVB rise and the RIB and AIB deltas lie inside the wild-type-only split-half range, so they are not separable from sampling noise; see Table 7) above it.

Across the 14 windows the share is almost perfectly anti-correlated with how many channels those are (Spearman -0.925 , $p = 2 \times 10^{-6}$), and seizures reduce that count sharply: over the six matched records, 11.7 driven channels per ictal window against 18.5 interictal, falling in five of six (one-sided $p = 0.047$). Total detectable drive does not change detectably in either direction: it falls in three of six records, its median rises ($0.47 \text{ to } 0.61$) while its mean falls ($1.15 \text{ to } 0.61$) on the strength of a single interictal window eight times larger than any other, and the paired test gives $p = 0.84$. Three statistics the clamp cannot inflate all show no ictal rise: the absolute drive carried by the top four channels (0.398 vs. 0.402 , $p = 0.42$), the concentration of $|\text{excess}|$ with no channel discarded (0.495 vs. 0.463 , $p = 0.42$), and the share normalised by the number of surviving channels ($p = 0.84$). In all six records the share moves opposite to the channel count.

So seizures do not recruit this network into a driven regime; they concentrate drive onto fewer channels without measurably increasing or decreasing its total. The measurement is real and the conditions are separable, which is what the prediction asked. What it is not is evidence of recruitment, and we had originally written it as though it were (Figure 9).

The third pre-registered prediction failed, and this failure is itself informative: per-event driven patterns do not replicate across the same patient’s seizures (mean pairwise Spearman 0.08 over 15 pairs). Whatever the cause—genuine per-event variability or single-window noise—the practical warning stands: single-event connectivity maps at these window lengths should not be treated as patient signatures without per-patient replication; we tested one subject.

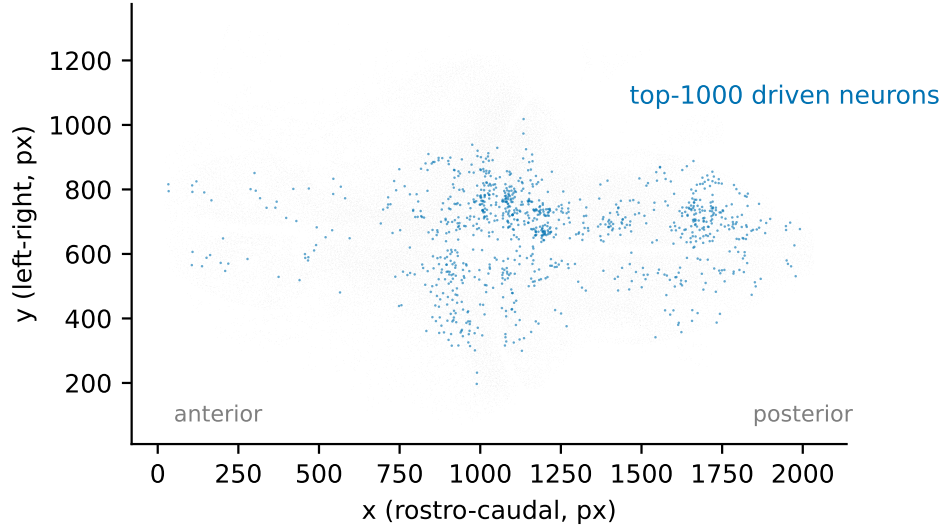


Figure 8: **The driven core of the larval zebrafish brain is spatially localised.** All 71,721 segmented neurons in horizontal projection (gray); the 1,000 highest-excess neurons (blue) concentrate posteriorly and near the midline (median nearest-neighbour distance 16.8 versus 29.8 ± 0.6 for random draws of 1,000).

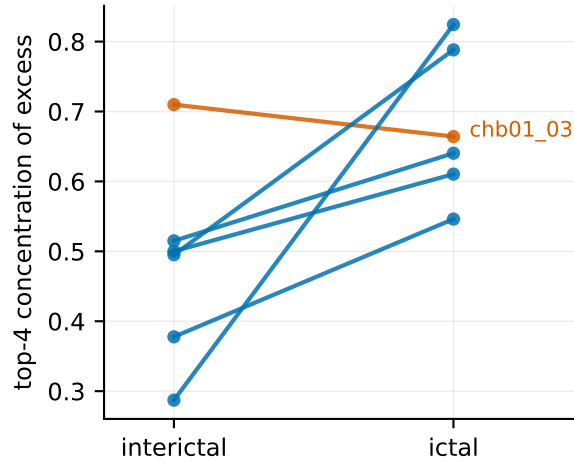


Figure 9: **Seizures concentrate drive onto fewer channels.** Top-4 channel share of total clamped-positive excess, interictal versus ictal, per record with an annotated seizure. Five of six records move in the predicted direction (paired Wilcoxon $p = 0.031$); the exception (record 03, red) is included. The share is a ratio over surviving channels, and seizures reduce the number of channels carrying detectable drive from 18.5 to 11.7 ($p = 0.047$) with no detectable change in total drive (median $0.47 \rightarrow 0.61$, $p = 0.84$), so this panel shows redistribution onto fewer channels rather than added drive; clamp-free versions of the statistic show no ictal rise ($p = 0.42$).

8.4 Climate: a tropical driven core and an invisible solar forcing

The three biological deployments share one inference target: a single organism’s internal organisation. The atmosphere supplies what none of them can: a driver that is exogenous by physics rather than by construction, which can be dropped in among ten thousand real variables and must score nothing.

Seventy-seven years of daily sea-level pressure (NCEP/NCAR Reanalysis-1; 10,512 cells, $\sim 28,000$ days) [18], deseasonalised and differenced, with the SILSO daily sunspot number [20] embedded as an additional channel. Pre-registered gates: ghost -0.005 ; **sunspot channel** -0.003 , a real exogenous forcing, embedded among ten thousand atmospheric variables, scores nothing: the cleanest field test of the statistic’s designed insensitivity to exogenous sources; and the driven core is spatially clustered (top-500 median nearest-neighbour distance 0.0435 vs. 0.0676 ± 0.0048 under a spatial null). The exploratory map, fixed to carry no prior claim, reproduces well-established large-scale structure: 99% of the 500 most-driven cells lie within 30° of the equator (mean $|\text{lat}|$ 6.1° vs. 45.6° for the grid). This is the Walker/ENSO belt [39, 40], where tropical pressure is subordinate to planetary-scale circulation [40], while polar and storm-track cells, dominated by locally generated baroclinic dynamics [27], rank most autonomous. Every cell shows positive excess; a single connected fluid contains no autonomous members (Figure 10).

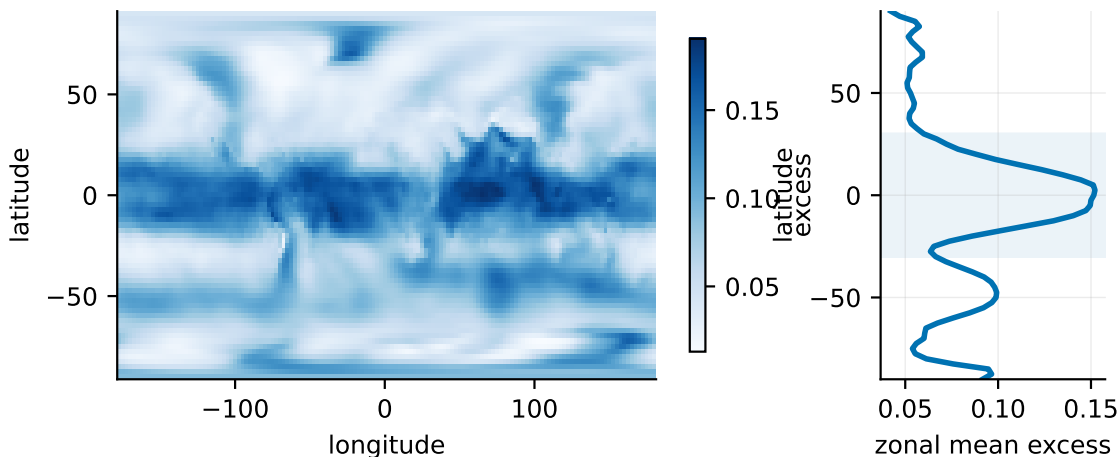


Figure 10: **The atmosphere’s driven core is the deep tropics.** Left: excess per grid cell for 77 years of daily sea-level pressure (single-hue scale; darker = more driven). Right: zonal mean excess by latitude; the shaded band marks $\pm 30^\circ$, which contains 99% of the 500 most-driven cells. Polar and mid-latitude storm-track cells—dominated by locally generated dynamics—rank most autonomous.

What the four establish together. No one of these deployments is decisive, and they are not four repetitions of the same demonstration. Two of them converge: two nervous systems, imaged by different techniques at a five-hundred-fold difference in width and analysed without labels in either case, place the driven core on the same class of machinery: the canonical motor-command ensemble in the worm, the posterior-midline territory associated with internally generated motor dynamics in the fish. Three of them test the same designed blindness, by unrelated natural

means: a hub silenced genetically (AVA, percentile $0.92 \rightarrow 0.38$ —an approximate source, and it falls accordingly), a forcing that is exogenous by physics (the sunspot channel, -0.003 among 10,512 atmospheric variables); the synthetic AR(1) family is the same test under construction. Blindness to sources is what makes the statistic falsifiable rather than merely descriptive, and it is the property most likely to be wrong; it survived two independent natural attempts to break it. A third case, the seizure-onset zone, is a source by clinical definition and should therefore be invisible, but the recording carries an event annotation and no channel-level ground truth, so we could not test it and do not count it. The deployments span V from 23 to 71,721 and n from 2,000 to $\sim 28,000$, and their ghost values are given individually in Table 6 rather than summarised. Two of them do not clear the sign rule of Section 5.5 and we name both. Wild-type worm 0 has a positive ghost ($+0.0035$); we retain it because it is the weakest of the three wild-type animals on every other measure (top-10 precision 8/10, against 10/10 for the two that carry the result), so excluding it would strengthen the reported findings rather than weaken them. The EEG deployment has five positive ghosts in fourteen windows, the largest $+0.022$, six times worm 0’s; restricting to the three records whose windows are both clean leaves the concentration effect non-significant ($p = 0.25$ against $p = 0.031$), and we report both. Readers who prefer the rule applied strictly should read the wild-type result as resting on worms 1 and 2 plus the two held-out animals, and treat the EEG concentration result as unreplicated. The EEG null marks the boundary this makes visible: the statistic resolves a change in network organisation between conditions well before it can be trusted to map a single event.

9 An empirical method-selection map

Because the goal is a usable tool, we summarise in Table 8 the regimes in which each method performed reliably in our measurements, including where ours should not be used.

The rows below draw on measurements from three places: the wall-clock and sample-complexity results stated in the Introduction, the synthetic systems of Section 7 and the near-synchrony chains run alongside them, and two datasets that appear only here: a 48-point circadian gene-circuit series [19] and the causal-chamber wind-tunnel system [21]. The latter two are too short and too far from the membership question to warrant deployments of their own, but they are where the short-series and near-linear recommendations come from; both are reported in full in the released ledger.

Absolute expectations matter as much as rankings: on real data the best methods deliver roughly $1.5\text{--}2.4\times$ baseline average precision, far from clean recovery.

10 Applying MACE to a new dataset

A method is only useful if a reader can run it on their own recording and know whether the answer is trustworthy. This section is the procedure we would follow on unfamiliar data, with the choices that matter, the diagnostics that catch the failures, and what to do when one fires. Every check below is computable without ground truth, which is the property that makes the procedure usable at all.

Table 8: **An empirical method-selection map.** Regimes in which each method performed reliably in our measurements, including where ours should not be used. Every recommendation is tied to a measurement reported in this paper or in the released ledger.

regime	recommendation	basis (measured)
pairs / small systems, weak coupling	CCM	detects coupling 0.01 in seconds; learned methods need 0.04–0.08
short real series ($n \lesssim 50$)	conditional k NN first (best AP, $1.75\times$ baseline); CCM and PCMCI beside it	measured on gene-circuit data [19]; no trained estimator viable
noise-driven, near-linear systems	PCMCI or CCM	statistically indistinguishable on wind-tunnel physics [21] (AP 0.265 vs. 0.256); learned reconstruction methods fail there and emit false positives on exogenous actuators
strong coupling near synchrony	conditional learned models (<i>candidate only</i> : synthetic chains, 3 seeds, unverified; did not transfer to the one dense real circuit tested)	CCM, conditional k NN and PCMCI all acquire mediated false positives from coupling 0.3–0.5 on synthetic chains
membership at $V \gtrsim 10^3$	this statistic	pairwise CCM is infeasible in our wall-clock measurements and k NN conditioning is sample-starved; PCMCI at this width is untested; top- k precision with ghost control
edge-level maps on dense real circuits	no method we tested	difference-based scores collapse under redundancy (Mechanism 1)

10.1 Procedure

1. Establish stationarity. Standardise each variable, and difference if there is drift. This is a prerequisite rather than a preference: under drift the circular-shift ghost inflates and the scan must be discarded rather than thresholded.

2. Choose the embedding. We used $E = 3$ throughout, with τ fixed per dataset on physical grounds and declared before analysis (Section 6.2); we did not run a data-driven selection rule, and a reader may reasonably prefer one, such as the first minimum of the auto-mutual-information. What matters is not the specific value but the condition Proposition 1 needs, $E > 2d_q$, and that condition is checkable in the next step without knowing d_q .

3. Compute the self-baselines first, and read them. The second term of Eq. 1 costs one ridge solve per variable and is required anyway. Its distribution is the single most informative diagnostic in the whole procedure. On a deterministic system a substantial fraction of channels should have self- R^2 near 1. If almost none do, then either E is too small, the polynomial degree is too low, or the data are noise-dominated, and in all three cases Proposition 1 does not hold, so autonomous channels can acquire spurious positive excess.

We are in that regime ourselves on every real deployment reported here (worm median self- $R^2 \approx 0.21$, maximum 0.35), so the honest instruction is graded rather than absolute. If

saturation fails you may still run the statistic, but you inherit none of Section 5’s guarantees: the output is then an empirical ranking that has to be validated against something outside itself, as the deployments in Section 8 are validated against connectome labels, a genetic intervention and known atmospheric dynamics. What you must not do is present such a run as carrying the no-false-positive property, and you should expect the ghost to be noisier than the theory implies.

4. Choose the function class by raising it until the diagnostics stop moving. We use degree 2 where the dynamics are known to be quadratic and degree 3 on real data. A too-weak class is the failure mode that produced our worst historical result, so the conservative direction is upward, bounded by sample size.

5. Train the code. A masked autoencoder with a hard linear bottleneck of width b , on contiguous three-way splits with an embargo. Under-wide b costs recall but not precision (Proposition 2), so err large subject to compute, and confirm by checking that the reported set is stable across two or three widths.

6. Build the ghost panel, and draw donors only from self-predictable channels. This is the step our own earlier runs got wrong, and Section 5.5 explains why it matters: the ghost’s null is guaranteed only when its donor’s self-baseline saturates. Rank channels by their one-step self-predictability (we screen with a nearest-neighbour analogue forecast, which tracks the Eq. 1 self- R^2 closely and is cheaper), keep those above a high threshold (we use 0.99), and build K ghosts by circularly shifting them by offsets drawn from the middle half of the record. Each ghost costs exactly what one more variable costs, so K can be large; we use 200.

Where no channel clears the threshold, as on the worm recordings, the rule cannot be applied and neither can its guarantee. Take the most self-predictable channels available, use many of them, and treat the resulting panel as a descriptive noise scale rather than a calibrated null. The deployments in Section 8 predate this rule and each used a single uniformly drawn donor, which is the procedure we score at precision 0.66 on the synthetics; their ghost values are therefore reported individually in Table 6, so that a reader can judge each scan rather than trust a summary of them.

7. Threshold at the panel’s upper extreme, floored at zero. Report variable q as driven when

$$\text{excess}(q) > \max\left(0, \max_{j=1..K} \text{excess}(g_j)\right). \quad (4)$$

The floor at zero is not a fudge: Proposition 1 bounds an autonomous channel’s population excess by zero, so a panel maximum landing below zero reflects sampling rather than a genuinely looser null. Over the three distinct synthetic graphs this rule gives precision 1.00 at 29% recall of the driven set, and 0.95 at 28% when the heterogeneous re-analysis is pooled in, against 0.66 precision for the single-ghost rule the earlier runs used.

10.2 Diagnostics, and what each failure means

All five checks in Table 9 are computable with no ground truth. We recommend reporting them alongside any result.

10.3 What it costs

The encoder is trained once per system; each variable thereafter is two ridge solves. The scan of 71,721 zebrafish neurons completed in roughly 14 minutes on a single consumer laptop, and the

Table 9: **Diagnostics and what each failure means.** Every check is computable without ground truth, and we recommend reporting all five alongside any result.

symptom	likely cause	action
ghost panel median clearly above zero	non-stationarity; the shifted copy no longer obeys the same recursion	difference or detrend and re-run; if it persists, discard the scan rather than thresholding it
many ghosts positive <i>and</i> self- R^2 low almost everywhere	function class too weak, or the process is stochastic rather than deterministic	raise the polynomial degree; if nothing moves, the saturation premise fails and the statistic is not licensed on these data
top scores of the same order as the panel maximum	no drive detectable at this sample size	report the null; do not lower the threshold to obtain a result
a large negative tail across many channels	the code is an uninformative addition for those channels, typically stochastic ones	ignore it; the lower tail carries no information about drivenness (Section 5.6)
reported set changes materially with bottleneck width	b below the $2d+1$ requirement	increase b until the set stabilises
a known-exogenous variable scores high	split leakage or non-stationarity, not a discovery	audit the splits and the embargo before believing anything else in the scan

200-ghost panels used in Section 5.5 were computed on already-saved encoders in a few minutes per system. The practical implication is that neither the panel nor a bottleneck-width sweep is a luxury: both cost a small multiple of a single variable’s readout, and both are what convert a number into a result a reader can check.

11 Limitations

The statistic’s semantics are narrow by construction, and the evidence above has boundaries that no ghost channel can detect. We state both, roughly in the order in which they would bite a new user.

(1) Drivenness only; sources are invisible, and a source-detection complement is future work. This is now derived rather than asserted (Proposition 1), and confirmed: zero of 63 sources were flagged over three distinct generating graphs (zero of 82 including the heterogeneous re-analysis of one of them). (2) Top- k semantics, and low recall. Roughly one driven channel in four is recovered at precision 0.95 (Section 7.1), so a variable’s absence from the driven core is *not* evidence that it is autonomous, and no threshold makes the output a complete inventory. (3) Validity is established down to $n \approx 2,000$ samples; no learned variant we tested succeeded on a 48-point circadian dataset, and the intermediate regime is unmeasured. (4) Stationarity is an essential assumption. (4b) The ghost’s guarantee is conditional on its donor: a ghost built from a driven channel carries no null, and the runs reported here used a single such donor. The corrected procedure (Section 10) draws a panel from self-predictable donors only; re-scoring the archived encoders under it raises precision from 0.66 to 0.95, and the headline top- k results are unchanged. That correction has been applied to the synthetic systems only: no real deployment here satisfies the donor rule, and on the worm no channel can, so the real-data ghosts remain

single-donor artifact meters, and six of them are positive: wild-type worm 0 (+0.0035) and five of the fourteen EEG windows (largest +0.022). (5) Single recordings per condition in the biological analyses; the intervention contrast is five animals per cohort from one preparation; the EEG analysis covers one subject; absolute excess magnitudes on the worm are small (top-band 0.02–0.23 across cohorts); synthetic validation is confined to map-type generator families. (6) The self-model’s function class governs transfer; the ghost and known-root channels are the practical guard. (7) The statistic estimates a known causal quantity; the contribution is scale, controls, failure analysis and validation, not the concept.

12 Conclusions and future work

Three failure mechanisms explain why learned causal readouts break on redundant dynamical systems. The correction they imply, excess over self-predictability computed against a compressed code of everything else, turns chance-level detection into high-precision membership scans. It does so at widths where pairwise scanning is infeasible in our wall-clock measurements and local conditioning is sample-starved, and it behaves as theory predicts across worm, fish, clinical and geophysical data, including under a genetic intervention.

That closes the gap stated at the outset in both of its halves: variable-level drivenness is estimable at $V \gtrsim 10^3$ —demonstrated to 71,721 variables in minutes on a laptop—and every scan carries a control that travels with the data rather than with a benchmark, so a result on a system for which nobody has ground truth can still be falsified. The evidence is not only correlational: on the worm, a statistic computed with no access to the perturbation registered a targeted genetic silencing in directions written down before the data was opened, at the level of the affected module (one cell survives correction for multiplicity across the ten contrasted cells). What the statistic cannot do is name sources, and the same self-referenced construction that makes it scalable and self-checking is what makes it blind to them—a source-detection complement built on the same subtraction is the natural next step, and the ghost channel is already the instrument for validating it. The operational protocol that found every failure here (embedded ghosts, known-root channels, trajectory checkpointing, pre-registration, precision-recall against stated baselines, published negatives) is, we believe, as transferable as the statistic itself.

Data and code. All datasets are public and unrestricted (OSF, ZAPBench, PhysioNet, NOAA PSL, SILSO, NCBI GEO, causal-chamber releases, OpenWorm). All code, the complete experiment ledger including negative results and voided attempts, the pre-registration documents, and the scripts that generate every figure in this paper from the primary data are available at <https://github.com/AkandaAshraf/DeepFeatSelection>. Code, pre-registration documents, and the complete experiment ledger—including negative results and voided attempts—are released alongside this paper.

References

- [1] G. Sugihara et al. Detecting causality in complex ecosystems. *Science* 338:496–500, 2012.
- [2] F. Takens. Detecting strange attractors in turbulence. *Lecture Notes in Mathematics* 898:366–381, 1981.
- [3] T. Sauer, J. A. Yorke, and M. Casdagli. Embedology. *Journal of Statistical Physics* 65(3–4):579–616, 1991.
- [4] J. Runge et al. Detecting and quantifying causal associations in large nonlinear time series datasets. *Science Advances* 5:eaau4996, 2019.
- [5] C. W. J. Granger. Investigating causal relations by econometric models and cross-spectral methods. *Econometrica* 37:424–438, 1969.
- [6] T. Schreiber. Measuring information transfer. *Physical Review Letters* 85:461–464, 2000.
- [7] A. Tank, I. Covert, N. Foti, A. Shojaie, E. Fox. Neural Granger causality. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 44:4267–4279, 2022.
- [8] Y. Xu, S. Zhao, J. Song, R. Stewart, S. Ermon. A theory of usable information under computational constraints. *ICLR*, 2020.
- [9] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, R. Girshick. Masked autoencoders are scalable vision learners. *CVPR*, 2022.
- [10] S. Kato, H. S. Kaplan, T. Schrödel, S. Skora, T. H. Lindsay, E. Yemini, S. Lockery, M. Zimmer. Global brain dynamics embed the motor command sequence of *Caenorhabditis elegans*. *Cell* 163:656–669, 2015.
- [11] J. G. White, E. Southgate, J. N. Thomson, S. Brenner. The structure of the nervous system of the nematode *C. elegans*. *Philosophical Transactions of the Royal Society B* 314:1–340, 1986.
- [12] S. J. Cook et al. Whole-animal connectomes of both *Caenorhabditis elegans* sexes. *Nature* 571:63–71, 2019.
- [13] J. M. Gray, J. J. Hill, C. I. Bargmann. A circuit for navigation in *Caenorhabditis elegans*. *PNAS* 102:3184–3191, 2005.
- [14] A. Sordillo, C. I. Bargmann. Behavioral control by depolarized and hyperpolarized states of an integrating neuron. *eLife* 10:e67723, 2021.
- [15] J.-M. Lueckmann et al. ZAPBench: a benchmark for whole-brain activity prediction in zebrafish. *ICLR*, 2025.
- [16] A. Shoeb. Application of machine learning to epileptic seizure onset detection and treatment. PhD thesis, MIT, 2009.
- [17] A. L. Goldberger et al. PhysioBank, PhysioToolkit, and PhysioNet: components of a new research resource for complex physiologic signals. *Circulation* 101:e215–e220, 2000.

- [18] E. Kalnay et al. The NCEP/NCAR 40-year reanalysis project. *Bulletin of the American Meteorological Society* 77:437–471, 1996.
- [19] M. E. Hughes et al. Harmonics of circadian gene transcription in mammals. *PLoS Genetics* 5:e1000442, 2009.
- [20] SILSO World Data Center. The international sunspot number. Royal Observatory of Belgium.
- [21] J. L. Gamella, J. Peters, P. Bühlmann. Causal chambers as a real-world physical testbed for AI methodology. *Nature Machine Intelligence* 7:107–118, 2025.
- [22] J. Theiler, S. Eubank, A. Longtin, B. Galdrikian, J. D. Farmer. Testing for nonlinearity in time series: the method of surrogate data. *Physica D* 58:77–94, 1992.
- [23] W. Ebisuzaki. A method to estimate the statistical significance of a correlation when the data are serially correlated. *Journal of Climate* 10:2147–2153, 1997.
- [24] T. W. Dunn, Y. Mu, S. Narayan, O. Randlett, E. A. Naumann, C.-T. Yang, A. F. Schier, J. Freeman, F. Engert, M. B. Ahrens. Brain-wide mapping of neural activity controlling zebrafish exploratory locomotion. *eLife* 5:e12741, 2016.
- [25] M. B. Ahrens, M. B. Orger, D. N. Robson, J. M. Li, P. J. Keller. Whole-brain functional imaging at cellular resolution using light-sheet microscopy. *Nature Methods* 10:413–420, 2013.
- [26] J. Runge et al. Inferring causation from time series in Earth system sciences. *Nature Communications* 10:2553, 2019.
- [27] E. K. M. Chang, S. Lee, K. L. Swanson. Storm track dynamics. *Journal of Climate* 15:2163–2183, 2002.
- [28] B. Lakshminarayanan, A. Pritzel, C. Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. *NeurIPS*, 2017.
- [29] L. Barnett, A. B. Barrett, A. K. Seth. Granger causality and transfer entropy are equivalent for Gaussian variables. *Physical Review Letters* 103:238701, 2009.
- [30] A. Kraskov, H. Stögbauer, P. Grassberger. Estimating mutual information. *Physical Review E* 69:066138, 2004.
- [31] A. Gordus, N. Pokala, S. Levy, S. W. Flavell, C. I. Bargmann. Feedback from network states generates variability in a probabilistic olfactory circuit. *Cell* 161(2):215–227, 2015.
- [32] K. Uzel, S. Kato, M. Zimmer. A set of hub neurons and non-local connectivity features support global brain dynamics in *C. elegans*. *Current Biology* 32(16):3443–3459, 2022.
- [33] W. M. Roberts, S. B. Augustine, K. J. Lawton, T. H. Lindsay, T. R. Thiele, E. A. Izquierdo, S. Faumont, R. A. Lindsay, M. C. Britton, N. Pokala, C. I. Bargmann, S. R. Lockery. A stochastic neuronal model predicts random search behaviors at multiple spatial scales in *C. elegans*. *eLife* 5:e12572, 2016.

- [34] N. Pokala, Q. Liu, A. Gordus, C. I. Bargmann. Inducible and titratable silencing of *C. elegans* neurons in vivo with histamine-gated chloride channels. *Proceedings of the National Academy of Sciences* 111(7):2770–2775, 2014.
- [35] C. Brennan, A. Proekt. A quantitative model of conserved macroscopic dynamics predicts future motor commands. *eLife* 8:e46814, 2019.
- [36] J. Meng, T. Ahamed, B. Yu, W. Hung, S. Ei Mouridi, Z. Wang, Y. Zhang, Q. Wen, T. Boulin, S. Gao, M. Zhen. A tonically active master neuron modulates mutually exclusive motor states at two timescales. *Science Advances* 10(15):eadk0002, 2024.
- [37] T. Kawano, M. D. Po, S. Gao, G. Leung, W. S. Ryu, M. Zhen. An imbalancing act: gap junctions reduce the backward motor circuit activity to bias *C. elegans* for forward locomotion. *Neuron* 72(4):572–586, 2011.
- [38] B. J. Piggott, J. Liu, Z. Feng, S. A. Wescott, X. Z. S. Xu. The neural circuits and synaptic mechanisms underlying motor initiation in *C. elegans*. *Cell* 147:922–933, 2011.
- [39] J. M. Wallace, D. S. Gutzler. Teleconnections in the geopotential height field during the Northern Hemisphere winter. *Monthly Weather Review* 109:784–812, 1981.
- [40] J. Bjerknes. Atmospheric teleconnections from the equatorial Pacific. *Monthly Weather Review* 97:163–172, 1969.
- [41] R. Vicente, M. Wibral, M. Lindner, G. Pipa. Transfer entropy—a model-free measure of effective connectivity for the neurosciences. *Journal of Computational Neuroscience* 30:45–67, 2011.
- [42] K. J. Friston. Functional and effective connectivity: a review. *Brain Connectivity* 1:13–36, 2011.
- [43] M. A. Kramer, S. S. Cash. Epilepsy as a disorder of cortical network organization. *The Neuroscientist* 18:360–372, 2012.
- [44] H. Ye, E. R. Deyle, L. J. Gilarranz, G. Sugihara. Distinguishing time-delayed causal interactions using convergent cross mapping. *Scientific Reports* 5:14750, 2015.
- [45] D. Janzing, B. Schölkopf. Causal inference using the algorithmic Markov condition. *IEEE Transactions on Information Theory* 56:5168–5194, 2010.
- [46] S. Löwe, D. Madras, R. Zemel, M. Welling. Amortized causal discovery: learning to infer causal graphs from time-series data. *CLear, PMLR* 177:509–525, 2022.
- [47] J. Devlin, M.-W. Chang, K. Lee, K. Toutanova. BERT: pre-training of deep bidirectional transformers for language understanding. *NAACL-HLT*, 2019.